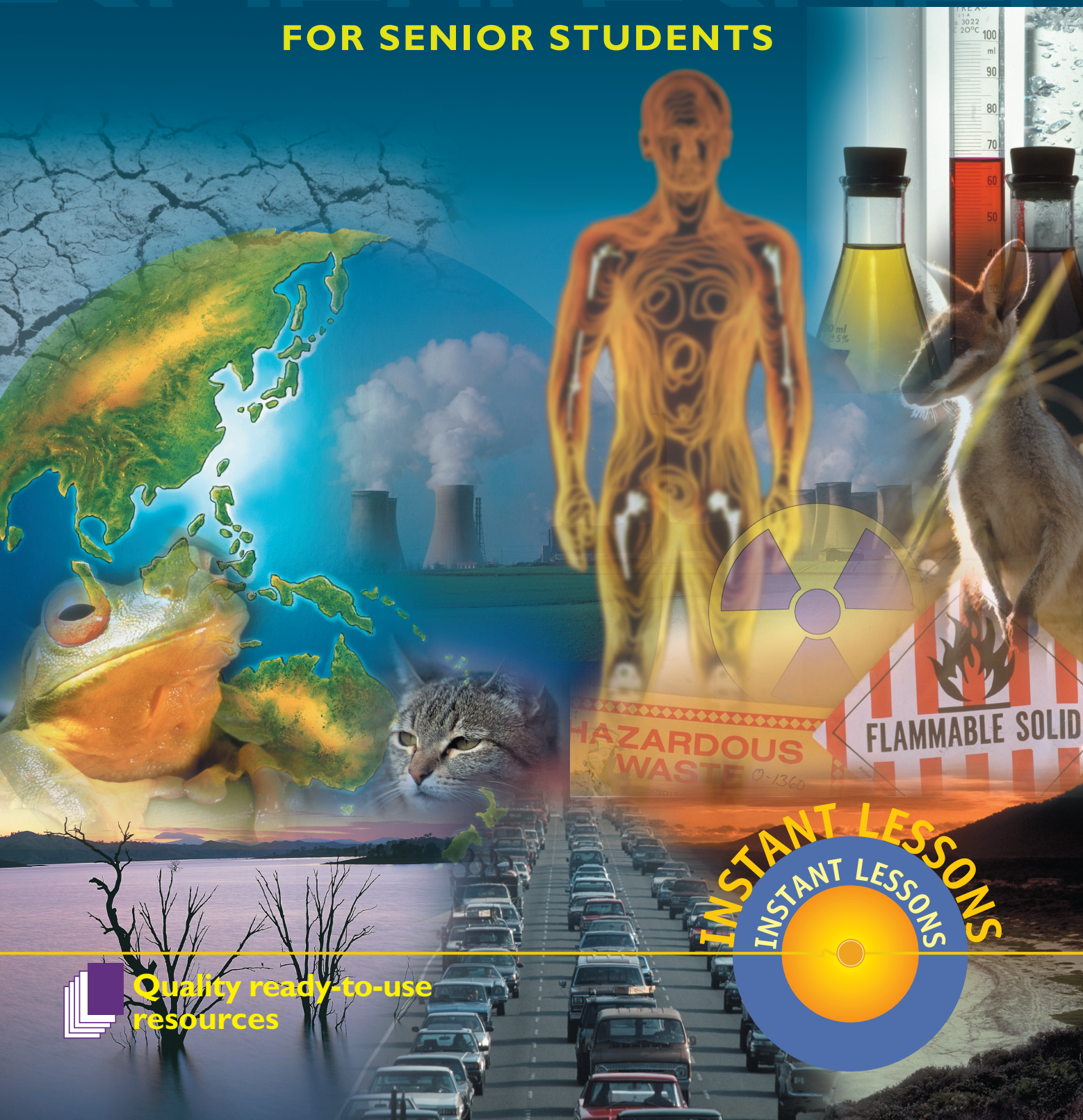


HOWARD CLARK and JOANNE HEAUME

1  
Book

# GENERAL SCIENCE TOPICS & ISSUES

FOR SENIOR STUDENTS



Quality ready-to-use  
resources

# GENERAL SCIENCE TOPICS & ISSUES

TOPICS FOR SENIOR STUDENTS

## BOOK 1

Howard Clark and Joanne Heaume



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Publisher: Susan Watson  
Editor: Shelley Barons  
Design and formatting: Infographics Pty Ltd  
Printed in Australia by Imprinta Pty Ltd



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An imprint of Learning Essentials  
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Ph: (03) 9548 0193  
[www.learningessentials.com.au](http://www.learningessentials.com.au)

ISBN 978 1 92561 058 1

# Introduction

**G**eneral Science Topics & Issues presents sets of guided questions and activities on a number of diverse Science topics for Years 9 to 12 students. It is designed to inspire students to extend their understanding, vocabulary and ideas beyond the classroom lessons.

The topics in this two-book series cover each of the major sciences (Biology, Chemistry, Geology and Physics) and may help Year 10 students to

choose their senior Science for matriculation by highlighting their particular areas of interest and expertise.

The questions range from basic recall through comprehension and interpretation, analysis and research, so as to extend the learning capabilities of students. Activities such as crosswords and group discussion encourage initial interest in the more formal activities.

## Contents

### 1 POLLUTED AUSTRALIA

|    |                                          |    |
|----|------------------------------------------|----|
| 1a | Environment Crossword                    | 4  |
| 1b | What is the Australian Environment Like? | 5  |
| 1c | How is Australia's Environment Changing? | 6  |
| 1d | What Makes Up the Environment?           | 7  |
| 1e | Environmental Change and Development     | 8  |
| 1f | Air Pollution                            | 9  |
| 1g | Air Pollution from Cars                  | 10 |
| 1h | Looking After Our Water Resources        | 11 |
| 1i | Water Pollution                          | 12 |
| 1j | Research Project                         | 13 |

### 2 AUSTRALIA'S ENVIRONMENT

|    |                                   |    |
|----|-----------------------------------|----|
| 2a | Australian Wildlife Crossword     | 14 |
| 2b | Coastal Degradation               | 15 |
| 2c | Habitat Destruction               | 16 |
| 2d | Sewage Pollution                  | 17 |
| 2e | Chemical Contamination            | 18 |
| 2f | Feral Animals                     | 19 |
| 2g | Feral Animal Control Program      | 20 |
| 2h | Global Warming                    | 21 |
| 2i | Greenhouse Effect                 | 22 |
| 2j | Research Project - Kyoto Protocol | 23 |

### 3 RADIATION AND SAFETY

|    |                                   |    |
|----|-----------------------------------|----|
| 3a | Radiation Crossword               | 24 |
| 3b | What is Radiation?                | 25 |
| 3c | Radiation and Health              | 26 |
| 3d | Ionising Radiation                | 27 |
| 3e | Continuing Nuclear Radiation      | 28 |
| 3f | Non-ionising Radiation            | 29 |
| 3g | Safety and Non-ionising Radiation | 30 |
| 3h | Extra-low Frequency               | 31 |
| 3i | Health Concerns and the Community | 32 |
| 3j | Research Project                  | 33 |

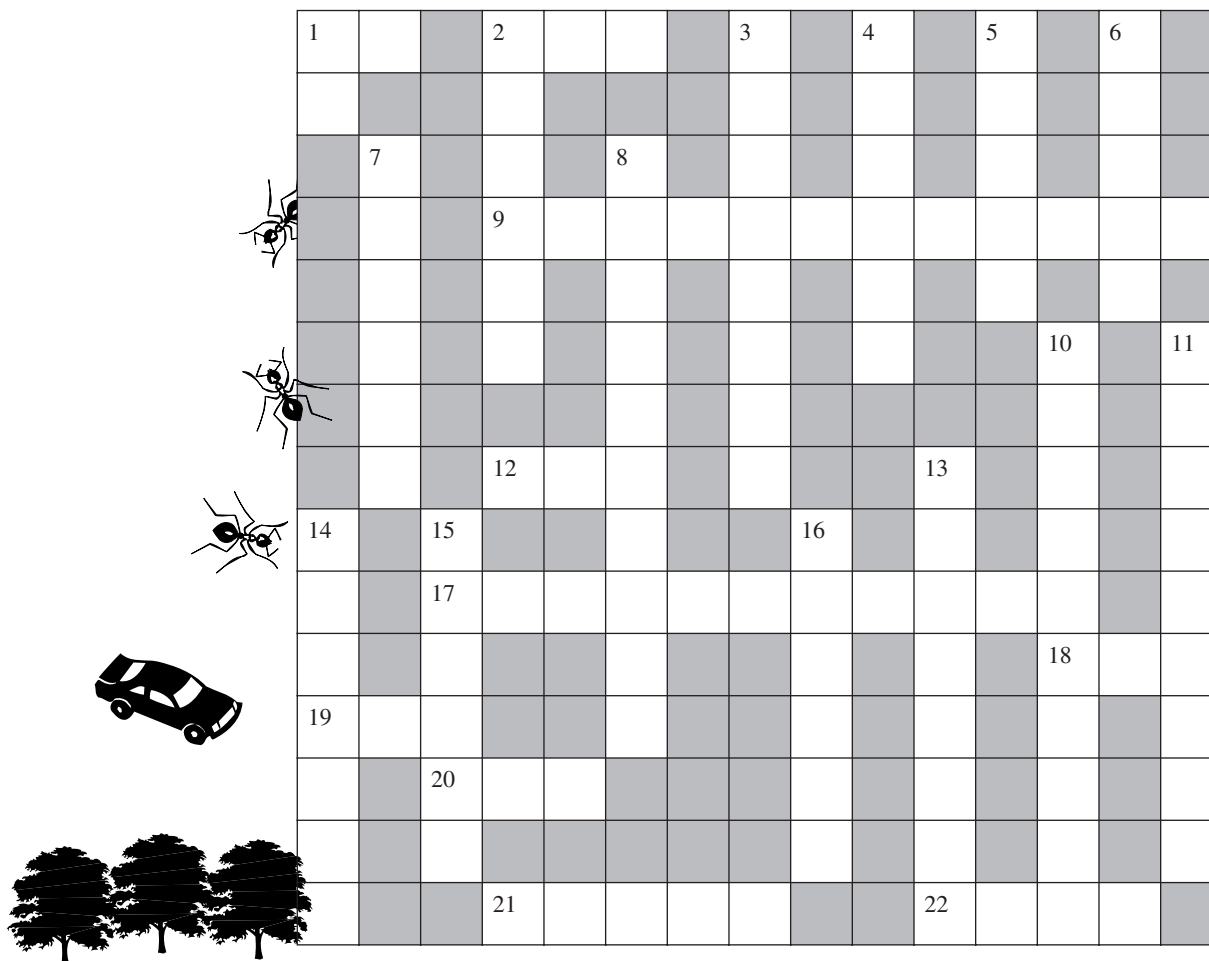
### 4 EVERYDAY CHEMISTRY

|    |                                     |    |
|----|-------------------------------------|----|
| 4a | Chemistry Crossword                 | 34 |
| 4b | Solutions, Suspensions and Colloids | 35 |
| 4c | Tyndall's Effect                    | 36 |
| 4d | Surfactants and Surface Tension     | 37 |
| 4e | Surfactants and Detergents          | 38 |
| 4f | Emulsions                           | 39 |
| 4g | Vitamins                            | 40 |
| 4h | Drugs                               | 41 |
| 4i | Research Project                    | 43 |

### ANSWERS 44

# Polluted Australia

## 1a Environment Crossword



### WORDS

ant  
biodiversity  
car  
chemical  
coast  
energy  
environment  
feral  
forest  
gas  
global  
Kyoto  
land  
native  
oil  
petrol  
pollution  
population  
salinity  
sea  
species  
UN  
US  
vegetation

### CLUES

#### Across

- 1 High-energy-using industrialised country (initials) (2)
- 2 Non-liquid, non-solid fuel used to produce energy (3)
- 9 Term relating to the number of different species of organisms in an area (12)
- 12 Water area of the continent (3)
- 17 General term for our living and non-living surroundings (11)
- 18 Small 6-legged animal that often gets trodden underfoot (3)
- 19 Object used for individual transport by many Australians (3)
- 20 Liquid fuel obtained from the Earth (3)
- 21 Term used to describe a domesticated animal that has gone wild (5)
- 22 Solid area of the Earth (4)

#### Down

- 1 Organisation responsible for establishing global environmental programs (initials) (2)
- 2 Term relating to the whole world (6)
- 3 The amount of salt in a river or creek (8)
- 4 Area containing trees, other plants and animals (6)
- 5 Edge of land in contact with the ocean (5)
- 6 Japanese city which gave its name to an environmental set of procedures (or protocol) (5)
- 7 Animal or plant originating naturally in Australia (6)
- 8 The number of plants, animals or people in an area (10)
- 10 Plant material (10)
- 11 Waste material which can destroy an environment (9)
- 13 General term for a substance containing atoms and molecules (8)
- 14 A distinct group of living things (6)
- 15 Liquid fuel burnt in many car engines (6)
- 16 Concept relating to heat and light released in fuels when burnt (6)

# Polluted Australia

## 1b What is the Australian Environment Like?

Today in Australia high levels of pollution are found in the air and water and concrete covers much of the ground in our cities. In arid (dry) regions we find erosion and salt pollution instead of forests and native bushland.

The Australia we know is far different from that found by the early European explorers just over 200 years ago. Those explorers and settlers started the cities that today we take for granted. Before that, however, in fact thousands of years ago, the first inhabitants of Australia started the slow changes to the environment.

- 1 What does the word 'environment' mean to you?
- 2
  - a List some types of pollution in Australia today.
  - b Do you or your family cause any pollution? If so, what type?
  - c Describe any problems caused by one of these types of pollution.

The Aboriginal peoples' long-term effect on the environment was minor as their population was small and spread out and they needed a healthy environment in order to survive. As the number of white settlers grew, the changes to the environment increased. The result is what we have today. We modern-day Australians sometimes forget that we also need a healthy environment for our survival.

- 3 Explain why prior to white settlement, the Aborigines only affected the environment very gradually.

- 4 List some differences between the Australia we know and the Australia found by the first European settlers.
- 5
  - a Most Australians today live in cities. Why do you think this is a problem for the environment?
  - b Draw a diagram showing how a settlement or a population centre such as a town or city affects the natural environment.

A number of years ago Melbourne was affected by a major dust storm. Many other towns and cities in Australia are regularly affected by dust. Dust storms are usually produced when dry winds blow the dry dirt from paddocks which have little or no grass. Often these storms happen during periods of drought. There were a number of dust storms during the 2002 and 2003 drought.

- 6
  - a What climate conditions produce dust storms?
  - b Describe the farming practices which may increase the possibility of dust storms.
  - c How could farmers reduce the effect of dust storms?
- 7
  - a Do you live in the city or country, near a beach, river or mountains?
  - b Do you think where you grow up influences your awareness of environmental issues?
  - c In the table below, list some ways city dwellers or country dwellers affect the environment. In the second column describe the effect.

| Effect of city/country dwelling | Effect on the environment |
|---------------------------------|---------------------------|
| i                               |                           |
| ii                              |                           |
| iii                             |                           |

# Polluted Australia

## 1c How is Australia's Environment Changing?

Australia has had many plants and animals from other countries introduced to the country over the years. Some of these include rabbits, common sparrows, even cats and dogs. We have increased our use of plastics, natural resources and energy. These changes are happening at an ever-increasing pace. Population increase means increasing needs for energy, technology and resources. It also means a rise in the production of wastes. This means that Australia today is changing much faster than the environment can respond.

**3** Have you noticed any examples of the problems described above in your own environment? If so:

- Describe the example you have seen.
- Imagine you are a researcher asked to return the situation to normal. What would you do? Why?

Research is being carried out to help us deal with these pests. The Pest Animal Control Cooperative Research Centre (PAC CRC) in association with the CSIRO is one such collaborative research area. It aims to develop bacterial and viral systems for delivery of fertility control for existing vertebrate pests (rabbits, foxes and mice) and to develop technologies for producing animals that are fertile only in captivity in order to prevent new feral pests.

The table below lists some ways introduced pests can affect our environment.

| Nature of pest              | Description of environmental effect                                                                                                                                              | Example                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <i>Predator</i>             | Eating many small native animals that are not used to having to protect themselves.                                                                                              | Feral dogs, cats and foxes         |
| Affecting food source       | Changing the number and type of animals available as food or protection for <i>parasites</i> such as ticks, fleas and other organisms that live on the bodies of larger animals. | Foxes and rabbits                  |
| Poisonous <i>prey</i>       | Killing or harming predators that bite or try to eat them by releasing poisons.                                                                                                  | Cane toad                          |
| <i>Competition</i> for food | Eating grass or plant material limits the food source in the diet of the native animals.                                                                                         | Sheep and rabbits                  |
| Affecting the land          | Eating grass down to the dirt line and digging burrows which varies the water runoff and helps in the <i>erosion</i> process.                                                    | Rabbits                            |
| Affecting plant life        | Taking over an area from slower growing native plants, cutting out the light and using the <i>resources</i> in the soil.                                                         | Lantana and other introduced weeds |

- What does 'fertility' mean?
  - How will making animals infertile change the environment?
  - In small groups, discuss possible ways could scientists make fertile animals infertile if the animals escape from captivity?
  - Decide in your group if these methods would be a good thing or would cause further problems. Present your argument in a report to the researchers.
- How do we find out about changes in our environment?
  - How do you think the early settlers and farmers found out about how they affected the environment?
  - What changes might the original Aboriginal inhabitants of an area notice in their environment?



**1** There are a number of *italicised* words in the table above.

- Write your own definition for each word.
- List some examples of each.

**2** List two introduced animals or plants in the table below. In the second column write some problems these animals or plants might cause to the natural environment.

| Animal or Plant | Effect on the environment |
|-----------------|---------------------------|
| a               |                           |
| b               |                           |

# Polluted Australia

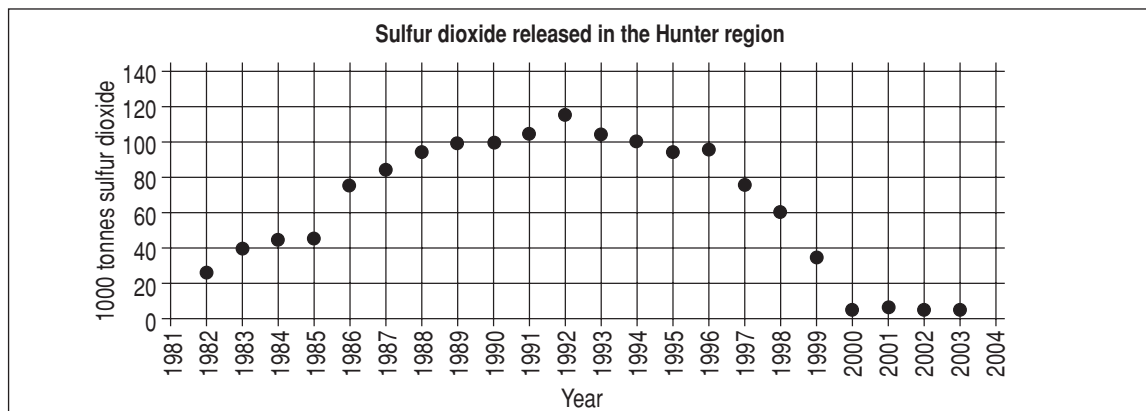
## 1d What Makes Up the Environment?

Researchers consider that there are ten major components of the Australian environment that need to be dealt with to slow the environmental destruction of our country. These components are: air pollution, biodiversity, coastal degradation, feral animals and plants, greenhouse gas production, native vegetation, population growth, river systems, salinity and urbanisation. The aim, of course, is to stop or even reverse this destructive trend. Change in any one of these areas will affect another. They are also affected by the environmental costs most of us are willing to pay to maintain our modern lifestyle.

- 1**
  - a What does the word 'component' mean in the paragraph above?
  - b Describe what you understand the ten major components to mean:
    - air pollution
    - biodiversity
    - coastal degradation
    - feral animals and plants
    - greenhouse gas production
    - native vegetation
    - population growth
    - river systems
    - salinity
    - urbanisation
  - c Why does change in one of these components affect the others?
- 2**
  - a How are the major Australian environment components different from 300 years ago?
  - b Draw a diagram showing one part of the environment 'then' and 'now'.
  - c Describe what caused any differences between the two diagrams.

The graph below shows information on one small part of the environment; the amount of sulfur dioxide ( $\text{SO}_2$ ) gas released in the Hunter region of NSW from 1982 to 2003.

- 3** Use the graph to answer the following questions.
  - a How many tonnes of sulfur dioxide were emitted in 1983?
  - b How many tonnes of  $\text{SO}_2$  were released in 1993?
  - c How many tonnes of  $\text{SO}_2$  were released in 2003?
  - d Describe any trend or overall change in the amount of  $\text{SO}_2$  released during the time period.
  - e State what you think might have happened in the years since 1983 and explain why.
- 4**
  - a What process or activity in the Hunter region was the cause of  $\text{SO}_2$  emission?
  - b In October 1999 the steelworks in Newcastle was shut down. How might this affect the production of  $\text{SO}_2$ ?
  - c Less  $\text{SO}_2$  is considered a good thing. What are the problems associated with closing a steelworks?
  - d Imagine you are a worker at the steelworks with a family in the Hunter region. Describe how you feel about the loss of the steelworks regarding the possible reduction in pollution, but also the reduction in employment.
  - e List some feelings you may have about the need to maintain jobs in a community and the need to protect the quality of our air and water.



# Polluted Australia

## 1e Environmental Change and Development

- 1
  - a What are some everyday objects we use which were not available to tribal Aborigines or early settlers?
  - b What are some activities we can do today which were not possible for tribal Aborigines or early settlers?
  - c Using the answers to (a) and (b) above, compare our way of life with that of early settlers or tribal Aborigines.
  - d List and describe some differences between our impact on the environment and that of the early Aborigines?

Read the following paragraph and use it to answer the next question

The words 'progress', 'employment', 'financial growth' and 'economic development' are used by people who say that environmental concerns are overrated. These people wish to reduce environmental support and protection. The words are also used by the ignorant and greedy who only think about short-term goals without considering long-term costs or the environmental results of short-sighted economic decisions.

A developer should consider productivity over the long term, and use the environment to develop and improve employment prospects and economic progress. This is done by using the natural advantages of the environment and considering the true long-term costs rather than destroying the environment in the short term.

- 2
  - a The writer of this paragraph presents some strong views. Rewrite the argument in your own words.
  - b Present some arguments of your own to agree *or* disagree with views expressed above.
- 3 Someone starting a business usually considers things like cost of raw materials and transport.
  - a What long-term considerations should also be made for a manufacturing business in the city?
  - b What long-term considerations should also be made for an agricultural business in the country?

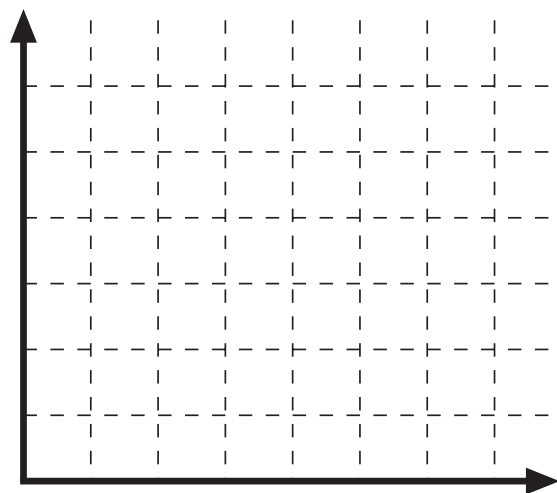
The effect on soil quality could be one consideration for an agricultural business. Soil acidity can be made worse by different farming methods. The table below shows soil samples, collected from areas of native vegetation and adjoining farmland in South East Queensland. Lower pH values (greater acidity) were obtained from soils taken from the cultivated area.

**Effect of farming on soil pH**

7 = neutral

| Depth cm | Soil pH of native vegetation area | Soil pH of cultivated area |
|----------|-----------------------------------|----------------------------|
| 0-10     | 5.5                               | 4.5                        |
| 10-20    | 5.2                               | 4.5                        |
| 20-30    | 5.0                               | 4.5                        |
| 30-50    | 4.9                               | 4.7                        |
| 50-70    | 4.6                               | 4.6                        |

- 4
  - a Label the graph then, using two different colours, graph the information from the table onto the grid.
  - b Suggest reasons why the farm soil pH might be more acidic than native vegetation soil.



- 5
  - a List some possible advantages or problems with farm soils being more acidic than native vegetation soils.
  - b What possible problems could there be if rainwater drops onto farmland then flows into creeks and rivers?

# Polluted Australia

## 1f Air Pollution

The combination of different concentrations of dust and chemicals in our air, particularly in our cities, causes breathing problems in both young and old people. Air pollution is also related to some respiratory diseases such as asthma. Air pollution comes from a number of sources including road traffic, factories, cigarette smoking and from our own homes.

- 1
  - a List two problems caused by air pollution.
  - b What are some major causes of air pollution?
  - c Describe the general quality of the air where you live. Does it contain any pollution? If so, describe the source of this pollution.
  - d What could be done to reduce the amount of air pollution in your area?

Until the 1980s, in Sydney the largest amount of dust in the air was produced by householders burning their own rubbish and dead leaves during autumn and winter. With the banning of backyard burning by many councils the amount of pollution from this source dropped dramatically.

- 2
  - a Sydney used to have a large dust problem. The graph below shows information on the atmospheric visibility in summer and winter. What is the difference between the coefficient of light in 1982 and 2002?
  - b Describe the trend shown by the graph.

- c What happens now to waste paper and autumn leaves (which used to be burned)?
- d Does your local council have an environmental policy on burning or waste removal? If so, describe the policy. If not, draft a letter to the council proposing that such a policy be introduced.

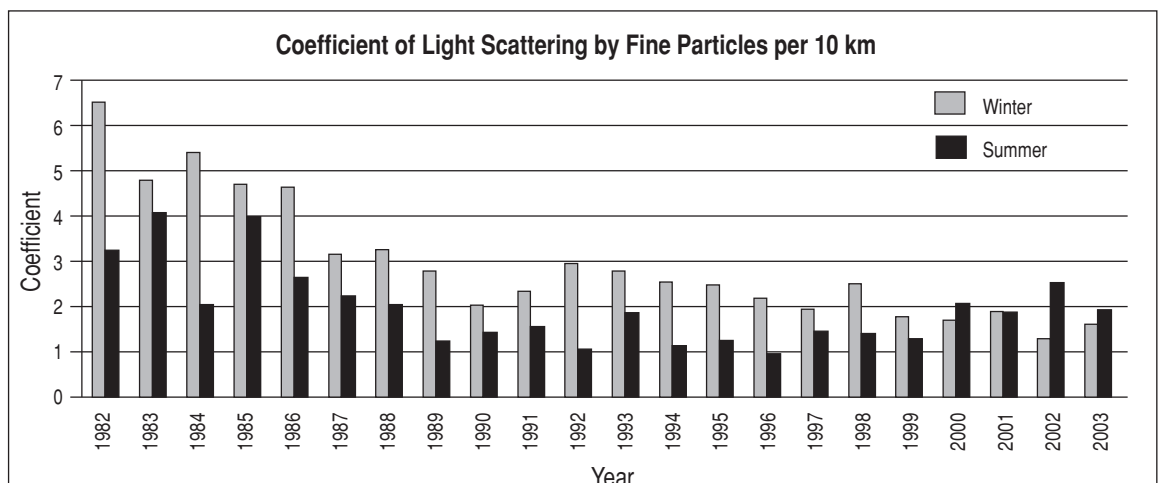
- 3
 

The graph shows the coefficient of light scattering by fine particles. This is how much of the light hits and bounces off fine particles in the atmosphere.

  - a What are two other terms we can use for 'fine particles'?
  - b Why does a higher level of scattering occur in winter than in summer?
  - c What trend is shown in the graph between 1982 and 1993?

Currently the biggest producer of air pollution in Australian cities is the internal combustion engine used in cars and trucks. Researchers are working on solar-powered cars or battery-powered electric cars or engines that use hydrogen to produce power, but at the moment they are very expensive to produce. The increasing number of cars combined with an inefficient, under-used public transport system produces a mixture of pollutants in the air.

- 4
  - a What causes most air pollution in cities today?
  - b Suggest ways this pollution can be reduced.



# Polluted Australia

## 1g Air Pollution from Cars

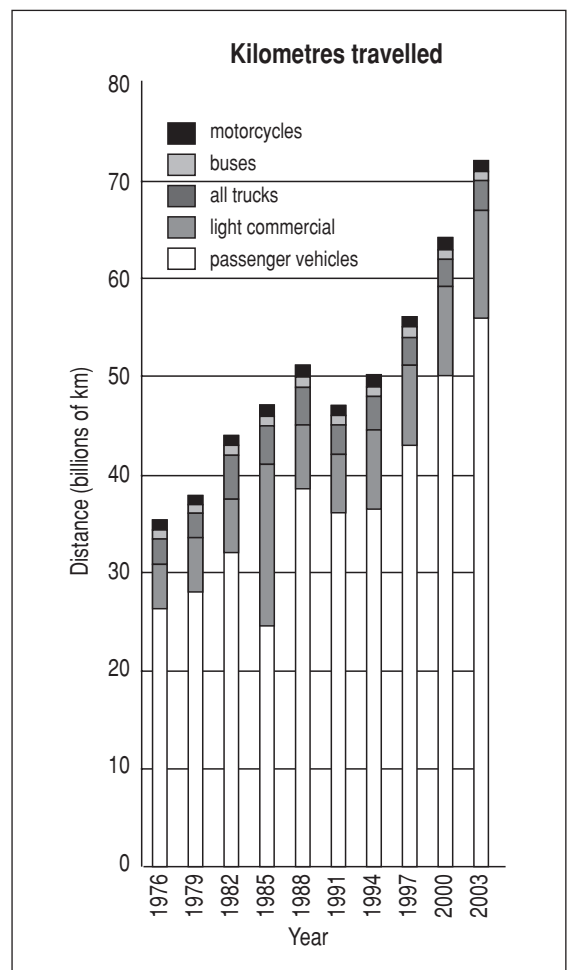
In all major cities commuters sometimes travel great distances to and from work. In Australian cities this commuting traffic is mainly to the centre of a city and so there is a concentration of roads, cars, trains and buses travelling to the city in the morning and away from the city in the evening. For those who live close to the city centre, public transport is frequent, inexpensive and convenient. The further away people live from the city centre, the more difficult it is to access public transport and the greater the use of private cars. People who do not work in the city centre are disadvantaged by lack of public transport. Some people regularly travel for up to 5 hours a day using different combinations of train, bus and ferry to get to work.



As our cities have grown, the development of large areas of urban sprawl on the edges of cities has led to an increase in air pollution. The cost of extensive public transport is large and would require a change in attitude by all who need to travel.

- 1
  - a How do you travel to school each day?
  - b List the advantages and disadvantages of using a car compared to public transport to travel to school.
- 2 The graph shows total kilometres travelled by different types of vehicles in Australia over several years.
  - a How many billion kilometres were travelled in 1988 and 1991?
  - b Why was there a decline after 1988?
  - c How many kilometres are driven by motorcycles each year?
- 3 In 2000 the 'best ever' Olympics were held in Sydney.
  - a How many kilometres were travelled in 2000 and in 2003?
  - b Predict the trend in passenger vehicles and commercial vehicles in 2005.

- 4
  - a How do you travel to entertainment and sporting venues on the weekend?
  - b List some advantages or disadvantages in walking to school or entertainment venues.
  - c Imagine you are conducting market research for public transport. Write a report for the managers of the public transport in your area on the usefulness of their services.
  - d If you were the manager of the local public transport providers, what response would you make?



# Polluted Australia

## 1h Looking After Our Water Resources

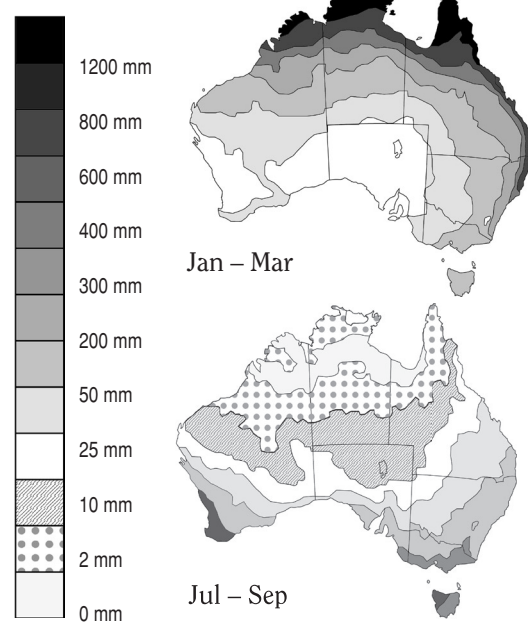
**W**ater pollution results from rubbish and waste released into our rivers and oceans. We do not notice the pollution until it interferes with our use and enjoyment of the water so, unless our drinking water is dirty or the water in the creek is covered with an oily scum or the beach we swim at has rubbish and detergent froth in the waves or dead fish are found in the lake, we do not believe the water is polluted. By that time the damage has been done.

Our Earth is unique in the Solar System because of its large amount of water - in the oceans, atmosphere, rivers and creeks and even the Arctic and Antarctic ice caps. Without water life could not exist on our planet. Most of the water is in the oceans but it is difficult to use because its high salt content. Fresh water is very limited and in some countries (including Australia) it is a scarce resource. Water therefore is very valuable and our use of water for drinking and washing, boating, fishing and farming can all be threatened by pollution. The survival of the animals and plants that live in water and on its edges, such as in tidal pools and river banks, also depends on minimising the level of pollution in the water.

- 1**
  - a List some of the ways you use clean water each day.
  - b What are some of the wastes that you or your family release into your household drains?
  - c What are some of the causes for the dirty water in our rivers or beaches?
  - d List some animals and plants that might be affected by pollution in your local river or creek.
- 2**
  - a Have you ever had to stop an activity involving water because the water was not clean?
  - b Describe the situation and suggest reasons for the water problem.
  - c Have you ever been responsible for littering or otherwise dirtying the water? Explain the situation. What should you have done?

According to the NSW Environmental Protection Authority, Australia is the driest continent and rainfall river flow volume is the most variable (least consistent) in the world. The maps below from the Bureau of Meteorology show the rainfall in Australia for two different seasons.

**Average Rainfall**



- 3**
  - a Use the map (or the Bureau website) and the key to find the average rainfall in your area for both January to March and July to September.
  - b Which months would be the dry season in the Northern Territory?

Often pollution collects on streets and in gutters until it rains, when it then passes through the stormwater drains and to a river or a beach.

- 4**
  - a How can we reduce the problem of pollution getting to our waterways?
  - b Three main sources of water pollution are cigarette butts, plastic bottles and oil. If you were in charge of the pollution control organisation for your area, how would you reduce the effect of one of these pollutants?

The NSW Environmental Protection Authority ([www.epa.nsw.gov.au](http://www.epa.nsw.gov.au)) and the Bureau of Meteorology ([www.bom.gov.au](http://www.bom.gov.au)) have information on climate and water pollution including interactive maps that allow you to find the rainfall in your area.

# Polluted Australia

## 1i Water Pollution

Our society is increasing its population continually, and therefore increasing its water usage. By the year 2030 the world's demand for fresh water will be greater than the supply. This problem is made worse by the fact that we are not reducing our pollution of the environment and we continue to waste our natural resources. Even today more than 30% of the world's population do not have regular access to fresh water. Every year about 15% of the world's population suffer illness from drinking polluted water.

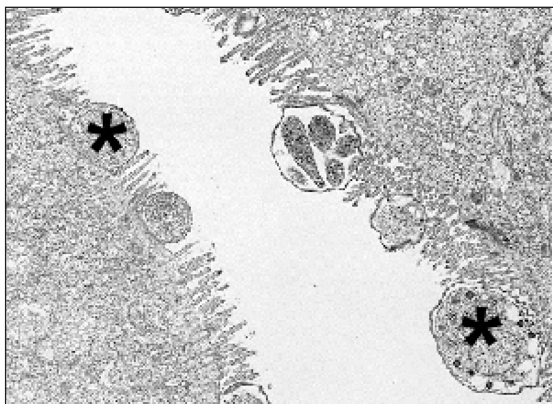
- 1 a How many glasses of water do you drink in a day?
- b Have you ever been sick from eating or drinking contaminated food or water? Imagine how it would be to regularly get sick from drinking dirty water. In a group, write how you might feel about this situation, and how you might prevent it.

In Australia, the quality of our water is generally very good. In Sydney the water is tested by NSW Health on behalf of Sydney Water. The results are published on the website and are available to consumers.

There are many forms of water pollution. Some of the common types are discussed below.

### A Microorganisms such as E coli and cryptosporidium

These microorganisms come from animal and human waste. The picture shows an electron micrograph of cryptosporidium (two are marked with asterisks) on the intestine of a sheep.



- 2 a What does the presence of harmful E coli in an animal's intestine indicate?
- b How could cryptosporidium get into our tap water?
- c This image is from a book about parasites. What is a parasite?

### B Chemicals used in industry that are poisonous to living things.

Heavy metals, such as mercury, are particularly dangerous because they accumulate in some of the fish and other seafood that we eat.

- 3 a What types of industries might use mercury, lead or other heavy metals?
- b How could these metals get into our water?
- c Why are these metals called 'heavy'? Aren't all metals heavy?

### C Oil

Oil floats on the surface of water and is therefore dangerous to birds as well as fish. Oil gets into our water from spillage or illegal dumping.

- 4 a Does your family use oil? Where is most oil found in the family home?
- b How could this oil get into our water?
- c Suggest why oil (which floats) is dangerous to fish that swim below the surface of the water?

### D Plastic

About 2000 plastic bags are used every year by the average Australian family. Most plastic pollution gets to the sea from suburban streets through stormwater drains.

- 5 a From where does your family get most plastic bags?
  - b How could you reduce the number of plastic bags you obtain?
  - c How can large plastic bags be dangerous in the sea?
- 6 a What advantages do you see in having an Environmental Protection Act in Australia?
  - b What changes might such an Act make to our way of life?
  - c What changes might such an Act make to our environment?

## 1j Research Project (group activity)

In this topic a number of websites have been listed. There are many other sources of environmental information. In particular, the websites for the various Environment Departments around Australia are very useful.

Bureau of Meteorology

<http://www.bom.gov.au>

Bureau of Rural Sciences (BRS)

<http://www.brs.gov.au/index2.html>

CSIRO Australia - Scientific and Industrial Research

<http://www.csiro.au/>

Environment Australia Home Page

<http://www.erin.gov.au/index.html>

Sydney Water

[www.sydneywater.com.au](http://www.sydneywater.com.au)

New South Wales National Parks and Wildlife Service

<http://www.npws.nsw.gov.au/index.html>

NSW Environmental Protection Authority

[www.epa.nsw.gov.au](http://www.epa.nsw.gov.au)

Northern Territory Parks and Wildlife Commission

<http://www.nt.gov.au/paw/>

Queensland Environmental Protection Agency

<http://www.env.qld.gov.au/>

South Australia Department for Environment and Heritage

<http://www.denr.sa.gov.au/index.html>

Tasmania Parks and Wildlife Service

<http://www.parks.tas.gov.au/tpws.html>

Victoria Department of Natural Resources and Environment

<http://www.nre.vic.gov.au/>

Western Australia: Welcome to NatureBase

<http://www.calm.wa.gov.au/>

Imagine you are an officer of the appropriate environmental or wildlife department from your state or territory and you and your team have been asked to prepare a report on the use of some land in your local area. The summary of your task is outlined below.

**Task** To determine the best use for an area of land taking into account environmental and commercial requirements of the region. Your group report will present the unanimous response and will be supported by correctly referenced resources. Your report may include visual aids (such as a computer slide show), verbal presentation and/or poster displays. You may support one (or a combination) of the following recommendations or develop your own response.

**Current land use** The area of land is currently in private ownership but 80% of it is covered by natural (untouched) vegetation typical of the area. It is the home to some mammals, birds and reptiles and unknown numbers of smaller animals. One species of bird and two species of frog are classified as 'threatened'. It borders on some urban land.

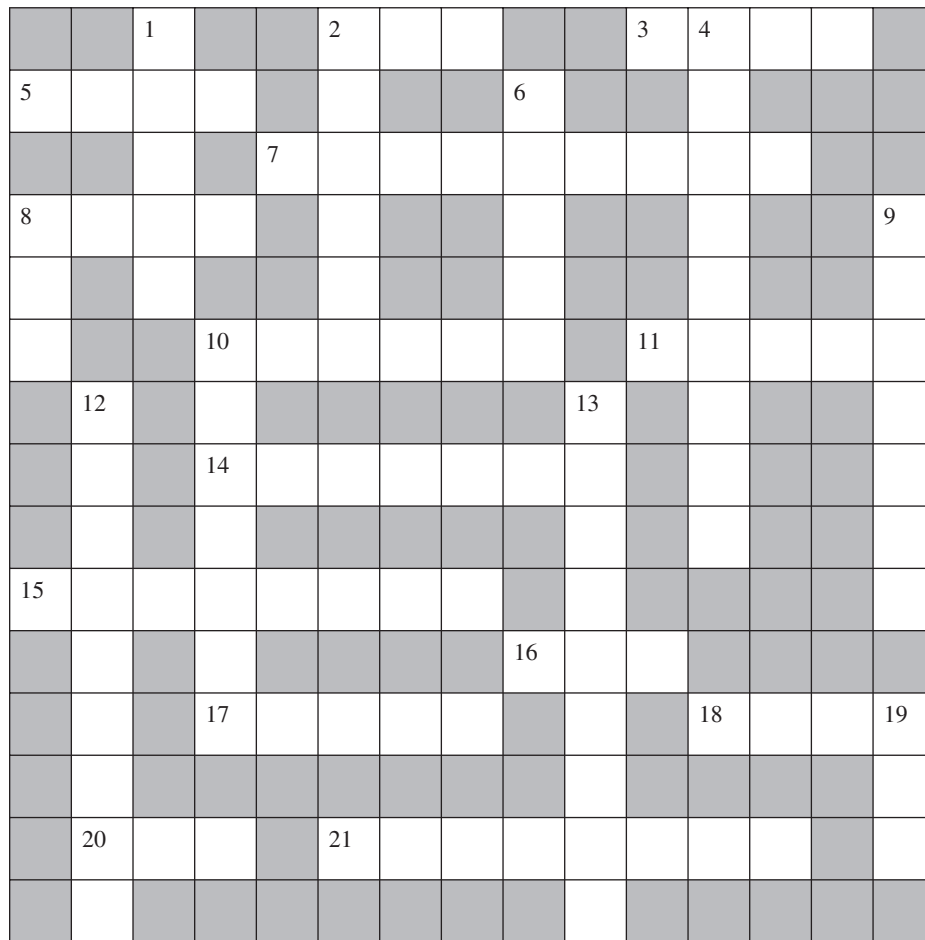
### Current recommendations

- 1 Fence off and leave untouched for future generations. Establish a breeding program for the endangered animals and study the area for unknown species. (*Green is the Future Colour Environmental Club*)
- 2 Establish a National Park with an education office and camp areas for students and visitors. Encourage family camping and develop a boardwalk through the area. (*Teach and Learn Association*)
- 3 Extend the neighbouring agricultural activities which will become more efficient through large scale mechanisation. This will enable the local area to gain much needed income. (*Farmers Union*)
- 4 Develop a manufacturing plant to encourage employment opportunities. The region's high level of unemployment needs a boost. (*Chamber of Commercial Entities*)

**Debate** If different groups in your class present alternative conclusions, present them in a debate to obtain levels of support for your idea within your class community.

# Australia's environment

## 2a Australian Wildlife Crossword



### WORDS

alligator  
back  
bat  
bill  
cod  
crocodile  
echidna  
emu  
frillneck  
frog  
galah  
gum  
heath  
kangaroo  
koala  
lyre  
marsupial  
mouse  
parrot  
paw  
platypus  
rosella  
waratah  
wattle

### CLUES

#### Across

- 2 Most animals have four of these and the state flower of WA is the kangaroo \_\_\_\_ (3)
- 3 One of our famous spiders is the red-\_\_\_\_ (4)
- 5 This amphibious animal is a good indicator of the environmental health of an area (4)
- 7 Large dangerous reptile which lives in estuaries and rivers in Northern Australia (9)
- 8 The platypus is sometimes called duck \_\_\_\_ (4)
- 10 The yellow coloured golden \_\_\_\_\_ is the floral emblem of Australia (6)
- 11 This common pink and grey bird is often found feeding on seeds beside the road (5)
- 14 Bright red coloured parrot found throughout the Eastern states (7)
- 15 It is a mammal but lays eggs, lives in burrows and feeds on small animals at the bottom of creeks (8)
- 16 Common name for eucalyptus trees (3)
- 17 This small shrub found in coastal areas is the state flower of Victoria (5)
- 18 This mimic has large tail feathers the same shape as an ancient musical instrument (4)

- 20 The Murray \_\_\_\_ is a huge freshwater fish now threatened (3)
- 21 The Latin name for many members of this animal is macropodus meaning large feet (8)

#### Down

- 1 Popular with tourists, this animal eats only one species of gum leaves (5)
- 2 General term for many species of Australian birds, including rosella, galahs and budgies (6)
- 4 Large reptile similar to and often confused with crocodiles (9)
- 6 The marsupial version of this animal is nocturnal and eats insects and worms (5)
- 8 Largely nocturnal flying mammal (3)
- 9 Spiny mammal that eats ants (7)
- 10 Large red state flower of NSW (7)
- 12 Descriptive name for a lizard with a flap of skin used when stressed (9)
- 13 Type of mammal whose young continue development in a pouch after birth (9)
- 19 Large flightless bird (3)

# Australia's environment

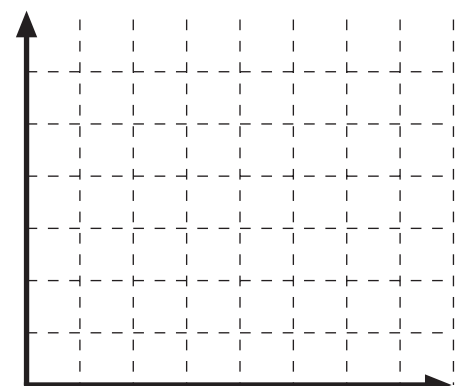
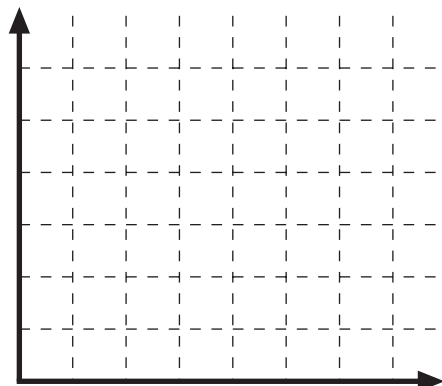
## 2b Coastal Degradation

Ninety per cent of the Australian population lives in the coastal regions of the continent and here the damage to the environment most affects the way people live. It is also here that the greatest changes to the environment have been noticed in recent years, with many people moving to live in areas which were previously uninhabited. Tourism, increasing urbanisation, the growth of industry and commercial fishing have all caused environmental changes on the coast. These changes are most noticeable when houses are suddenly under threat from very high tides or storm waves.

- 1
  - a Describe where the greatest concentration of people live in your state.
  - b What parts of the coastal region are affected by factories or houses?
  - c What does the term 'urbanisation' mean to you?
  - d Describe how the coast of your state is affected by factories or urbanisation. How can increasing urbanisation affect the coastal regions?
- 2
  - a Name one activity that causes changes to the coastal environment.
  - b What happens during this activity?
  - c Explain how this activity affects the environment.

The table shows the area of beach sand dunes around the states and the proportion of beach around the coastline.

| State | Area (km <sup>2</sup> ) | Beach (%) |
|-------|-------------------------|-----------|
| Qld   | 5110                    | 19        |
| NSW   | 1240                    | 16        |
| Vic   | 1650                    | 28        |
| Tas   | 980                     | 14        |
| SA    | 5610                    | 53        |
| WA    | 12 060                  | 33        |
| NT    | 1240                    | 6         |
| Aus   | 27 900                  | 24        |



- 3
  - a Which state has the smallest proportion of coastline beach?
  - b What is the area of the dune and beach coast in WA?
  - c What percentage of the coast of the most populated state is beach?
  - d Draw two column graphs showing the area and proportion of beach coastline for each of the states. Give each a title and label the axes.

The most obvious problem with beach erosion is the loss of sand caused by tidal movement and water currents. Coastal urban areas (where people live) are often potentially dangerous because landowners have built on or too close to the sand dunes themselves. What some people do not understand is that the wind and water moves sand on to and off the beach and because of this the sand dunes are always changing. Houses built on sand dunes run the risk of one day disappearing, taken by the eroding seawater. In some cases, building on the sand dunes also exposes the deeper sand not protected by grass, therefore making the sand easier to remove.

- 4
  - a What might happen during a storm or high tide if people have built their houses too close to the coast?
  - b What normally happens to a sand dune over time?
  - c How does this cycle affect houses built on sand?
  - d Why is the deep sand easier to erode?

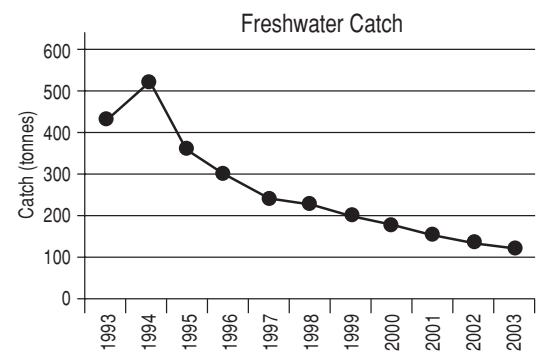
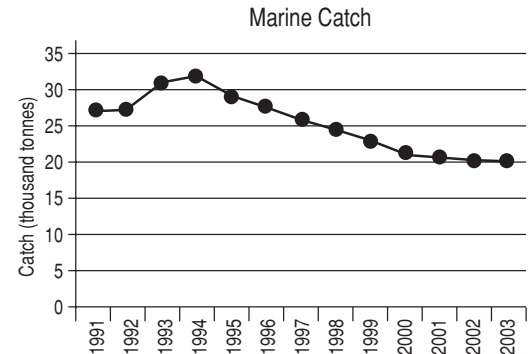
# Australia's environment

## 2c Habitat Destruction

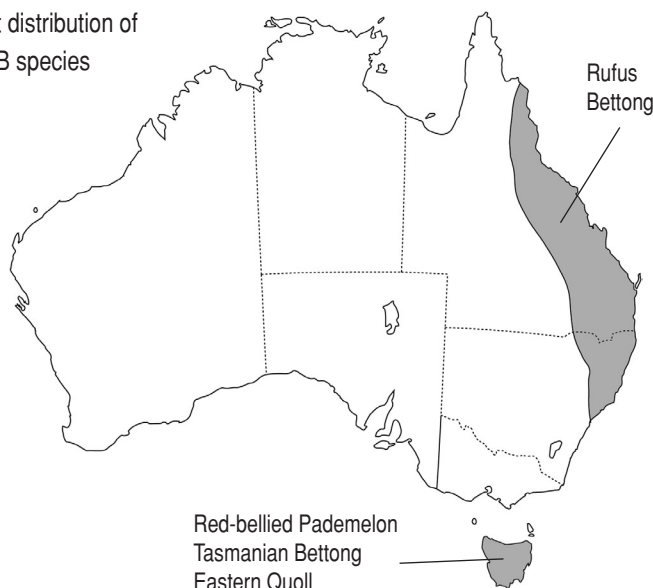
The map below shows the distribution of four Group B (ie considered threatened) marsupial species, including the Rufus Bettong. This species used to be common over the whole eastern part of Australia, but is now extinct in Victoria, probably due to predation (being eaten for food) by the Red Fox and feral cat. The destruction of the bettongs habitat and competition for food by rabbits, cattle and sheep also helped in their extinction. Bettongs can be found close to the coast, but also inland.

- 1
  - a What possible influence could coastal erosion have on the Rufus Bettong?
  - b How could urbanisation destroy the habitat of the bettong?
  - c Do you think the Rufus Bettong would be commonly found around small towns and farms in Queensland? Explain your answer.
  - d How could the distribution of this animal be increased?
- 2
  - a Mark on the map where you live.
  - b Is there a rare or threatened species in your local area? Can you name it?
  - c Describe what you or your community could do to help increase the population or distribution of a threatened species.

Sometimes an animal is killed for food or sport. The two graphs below show the total catch for marine and inland fisheries in NSW.



Current distribution of Group B species



- 3
  - a Which year had the highest catch of marine and freshwater fish?
  - b How many tonnes of marine (saltwater) fish were caught in 1996-97?
  - c How many tonnes of freshwater fish were caught in 1996-97?
  - d Suggest why there is such a huge difference between the two figures?
  - e What has happened to the annual catch of both types of fish over the last 8 years?
  - f Suggest some reasons why this trend has occurred in the waters of NSW.
- 4 Suggest some reasons why a continuation in the trends shown in the graphs above may pose problems for humans.

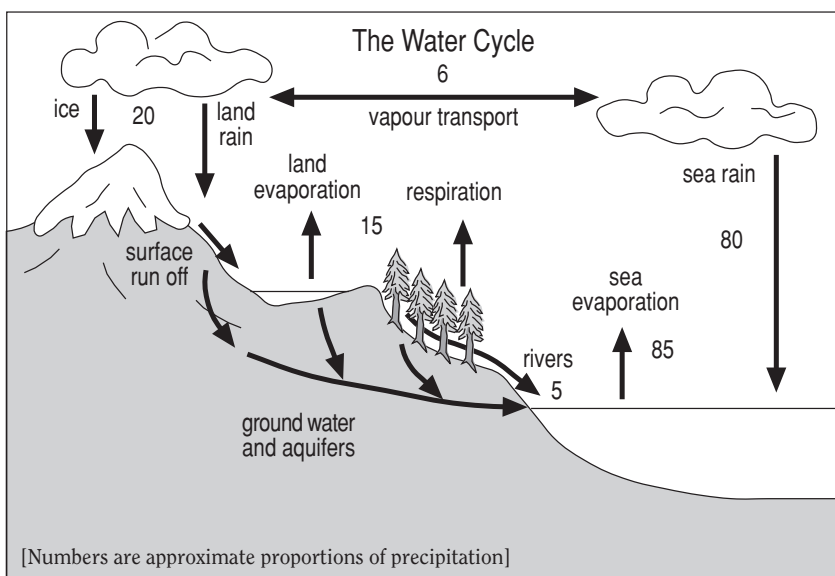
# Australia's environment

## 2d Sewage Pollution

Sewage is full of raw nutrients returning to the environment. If sewage is returned close to urban areas the chance of disease spreading is high. The release of sewage into rivers also increases the chance of algal blooms, which cause oxygen to be reduced. An algal bloom happens when there is so much food that the algae grow very quickly. This means the available food and energy for other living organisms disappears and the area dies. Slow natural breakdown of wastes is required so that nutrients can be safely returned to the environment. Biochemical Oxygen Demand (BOD) is a measure of the quantity of dissolved oxygen used by organic matter under specified conditions. BOD can be used to predict the effect waste may have on fish or other aquatic organisms that require oxygen to live.

- 1
  - a In your own words, what is Biochemical Oxygen Demand (BOD)?
  - b Is there any way you can stop pollution reaching the estuaries?
  - c What is the best way of reducing the effect of water pollution on the ecosystem?
  - d We all produce wastes. In small groups, determine a way of reducing the effect of our own sewage on the beaches, rivers and estuaries.

The diagram shows the movement of water around the coastal environment.



- 2
  - a Imagine you are a raindrop in the cloud ready to fall as rain. Describe what could happen to you as you pass through the different systems shown in the diagram.
  - b What proportion of water falls on land? What proportion falls over the sea? What proportion is lost from the land surface as evaporation and respiration? What proportion runs into the oceans through rivers?
  - c There are no urban areas in this diagram. If this area were settled, where would people live and where could sewage contamination occur?

There are a number of sewage treatment processes. One example is the system used by Ecomax, based in Western Australia. It is currently used in a number of states. Using gravity, the waste water gradually flows through a material called red mud. The material is called red mud because the main ingredient is alumina refining residue, which is high in iron and therefore red coloured. Chemical and biological processes remove phosphorus, nitrogen, ammonia, disease-causing organisms, suspended solids, heavy metals and many other constituents.

At Kwinana Sewage Treatment Plant in WA, the following results were obtained.

| Parameter (pollutant) | Final concentration (mg/L) | Removal efficiency |
|-----------------------|----------------------------|--------------------|
| phosphorus salts      | 0.01-0.05                  | >99.6%             |
| nitrogen salts        | 2.0-10.0                   | >80%               |
| ammonia               | <5                         | >90%               |
| BOD                   | <10                        | >90%               |
| faecal coliforms      | 0-500/100 mL               | >99.5%             |
| suspended solids      | <10                        | >90%               |

- 3
  - a Which two pollutants are most efficiently removed by this process?
  - b When are faecal coliforms found?
  - c Are they dangerous?

For further information on this process you can visit the website <http://www.ecomax.com.au/>.

# Australia's environment

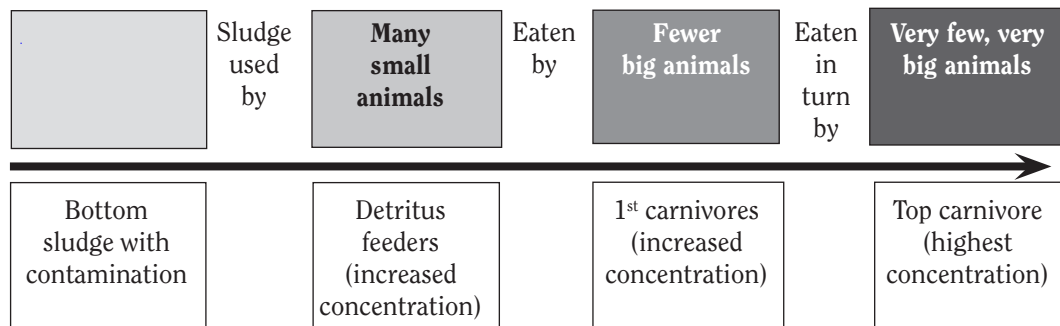
## 2e Chemical Contamination

### ESTUARY POLLUTION

As well as affecting coastlines and beachfronts, coastal degradation also happens in estuaries. An estuary is the part of the river near its mouth and is affected by the tidal movement of water. At high tide the sea water moves up the estuary and at low tide the fresh water moves out to sea. Pollution of the estuary is mainly due to sewage and stormwater runoff and can also include heavy metal pollution from industry. The pollution adds nutrients, pesticides, sediments, poisons and plastics to the drains, creeks and rivers and therefore to the water in estuaries. Contaminants often sink to the bottom of estuaries and remain for a long time in the sediment.

- 1 a Describe what an estuary is.
- b What happens during tide changes in an estuary?

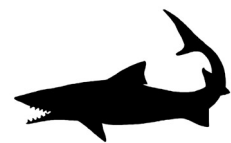
Animals such as prawns that live on the sediment take in pollutants as they feed. Bigger animals such as fish eat these small animals and the contamination becomes more concentrated. The last animal in the chain, the big carnivorous fish, are caught for human consumption and contain the poisons from all the small fish they have eaten. This can affect us.



- 2 a In what type of environment do prawns live?
- b What sort of animal is a detritus feeder? Give an example of a detritus feeder.
- c What does a carnivore eat? Give an example of a carnivore.

- 3 The boxes in the diagram above represent the number of animals and the shading represents the concentration of pollution.

- a What group has the most animals?
- b Which level has the fewest animals?
- c Name the group with the lowest concentration of pollution.
- d Where would we find the highest concentration of pollution?



- 4 a How does the concentration of pollutants increase in the big carnivores?
- b The diagrams below show examples of organisms that live in estuaries. Draw arrows between them to show the direction of food, energy and pollution.



- c Draw a diagram showing examples of which animals eat other animals in nature. This is called a food chain. An example (non-specific) might be *small fish* ⇒ (is eaten by) ⇒ *large fish*.

# Australia's environment

## 2f Feral animals

**F**eral animals are a major problem all around Australia and in particular the edges of the towns. This is where the natural environment is under greatest threat from humans. Many native animals including possums, birds, lizards, frogs and spiders are at threat from human pets and introduced or exotic animals and plants. When domestic pets escape or are abandoned they can turn feral and become very efficient killing machines in suburban bush areas. Some other examples of feral animals in Australia include: carp in the rivers, European green crabs off the southern coasts, wild pigs in the dry northern inland, foxes and deer in the eucalyptus forests and the most famous of them all, the rabbit.

**1** Describe what the following terms mean and give an example for each:

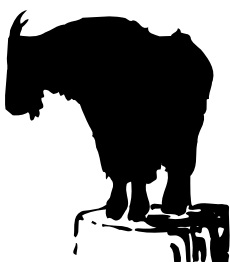
- urbanisation      • feral              • domestic
- eucalyptus        • predator      • carnivore
- suburban

Cats and dogs are well known for being carnivorous predators. When released into the wild, by accident or on purpose, these animals need to catch prey to survive and soon return to their instinctive hunting nature. The prey they can find mainly includes small native birds, frogs and lizards, but their diet can even include some species of kangaroo mice. Predation by feral pets is a major problem around the large cities.

- 2**
- a List some examples of native animals threatened by urbanisation.
  - b List some examples of introduced species in Australia.
  - c Choose one introduced species and describe what problems it may cause in our environment.

### FERAL GOATS

Australia has approximately 2.6 million feral goats and they can be found in all states and territories. Feral goats are both an agricultural and environmental pest, but they can also be useful as commercial stock. Feral goats compete with sheep and some native animals for grass, and help cause erosion due to their eating style. A number of ideas have been considered to solve the problem of feral goats.



- 3**
- a List some ways you might try to reduce the numbers of feral goats in the environment.
  - b Describe some possible problems or difficulties with your ideas.
  - c Would there be any arguments to support the retention of feral goat numbers? Describe them if there are and suggest responses.

The table below shows a summary of the initial densities, levels of reduction and costs of various goat control techniques used in a semi-arid environment. Each result is from a different study.

| Technique           | Density goats/km <sup>2</sup> | % Reduction | Cost per goat (\$) |
|---------------------|-------------------------------|-------------|--------------------|
| Helicopter shooting | 6.22                          | 45          | 3.95               |
|                     | 1.19                          | 70          | 7.16               |
|                     | 26.00                         | 85          | 7.30               |
|                     | 6.99                          | 99          | 15.00              |
|                     | 1.85                          | 13          | 19.70              |
|                     | 14.50                         | 49          | 2.91               |
|                     | 5.10                          | 75          | 4.62               |
| Mustering           | 4.75                          | 100         | 19.10              |
|                     | 2.73                          | 32          | 1.00               |
|                     | 23.00                         | 26          | 4.18               |
| Ground shooting     | 7.50                          | 26          | 3.50               |
|                     | 5.60                          | 0           | 774.00             |

Your job (as a senior Government Advisor) is to graph the material in the table to show the most efficient and cost effective way of solving the feral goat problem. Present your results in the form of a report.

- 4**
- a In a group, determine how to present the information in graphical form to best show how you arrived at your conclusions.
  - b What might be some other advantages of your chosen method?
  - c List some possible disadvantages of your chosen method.
  - d Think about the processes you used to arrive at your answers for parts (a), (b) and (c). Did you all have input or did one person determine the answers?

# Australia's environment

## 2g Feral Animals Control Program

The National Feral Animal Control Program (NFACP) was established by the Federal Government to develop and implement a program to reduce the damage to agriculture and the environment caused by pest animals. The NFACP is run by the Bureau of Rural Sciences and the Biodiversity Group of Environment Australia with the cooperation of State, Territory and Local Governments.

Pests are not limited to the well known goats and rabbits, cats and dogs. Many of you will know about the problems regarding cane toads, red algae and lantana.

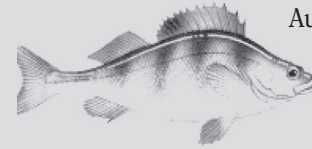
- 1 List as many problem or pest animals as you can and determine how they affect the natural environment. (Many Local Councils will have rangers whose job involves the removal of pest plants and the catching of pest animals.)

The table below shows the introduced species in various river systems. The introduced species are in *italics*.

| Regional community                                                                                        | Dominant species                                                                                                              |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Montane fish community (rivers above 700 m elevation)                                                     | mountain galaxias, eels, <i>brown trout</i> , <i>rainbow trout</i>                                                            |
| Lowland rivers of the North Coast                                                                         | long-finned eels, freshwater catfish, Australian smelt, Australian bass, freshwater mullet, empire gudgeons, striped gudgeons |
| South Coast fish community                                                                                | long-finned eels, Australian smelt, Australian bass, striped gudgeons                                                         |
| Inland rivers fish community (including the Darling River and tributaries)                                | <i>carp</i> , <i>western carp</i> , gudgeons, <i>gambusia</i> , bony herring, golden perch                                    |
| Murray River fish community (including the Murray, Murrumbidgee and Lachlan rivers and their tributaries) | <i>carp</i> , <i>rainbow trout</i> , Macquarie perch, golden perch, Australian smelt, <i>redfin perch</i>                     |

- 2
  - a From the information above, make a new table to list the native and introduced fish species.
  - b Some of the introduced fish are carnivores and others compete with the native fish for food. Describe how each type might cause a reduction in the number of native fish.

One example of an introduced fish is the **Redfin (English perch)**. Redfin has been introduced to Australia and is



thought to be responsible for the reduction in numbers of many

different native species. Anglers are instructed that redfin should not be returned to the water alive and that it is an offence to move redfin from one place to another. Regardless of their environmental problems, redfin are considered a good fighting fish and are therefore popular to catch. They are also a good fish to eat, which increases their popularity.

- 3
  - a What instruction do anglers receive regarding redfin perch?
  - b Why should the fish not be moved?
  - c What advantages are there in catching redfin?

The main objectives of the NFACP are to: 1) develop integrated, strategic approaches to manage the impacts of nationally significant pest animals; 2) improve the effectiveness of control techniques and strategies for reducing pest animal impact; 3) produce guidelines for the management of nationally significant pest animals; and 4) implement Threat Abatement Plans for managing pest animal species.

- 4
  - a What does NFACP stand for?
  - b What information would a Threat Abatement Plan include?
  - c The table above has information about fish types in river systems. Describe how this could be useful for a Threat Abatement Plan.
- 5
 

Group activity You are in charge of developing a new government initiative program to protect the Australian environment by controlling a feral animal or plant.

  - a Develop a name for the project and describe its objectives.
  - b What information would it collect and what rules would it put in place?
  - c How would we know if your program was successful?

# Australia's environment

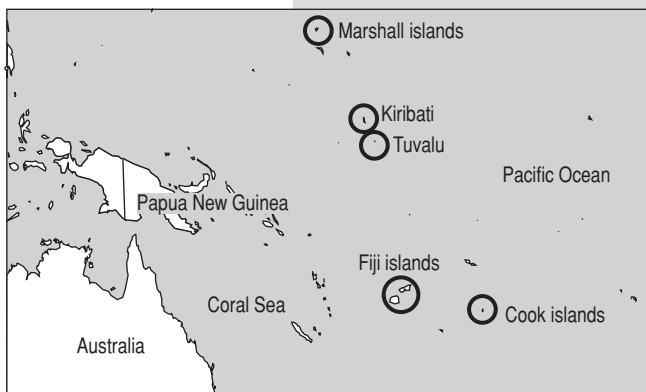
## 2h Global Warming

**G**lobal warming is expected to have a huge effect on our world over the next 5 to 10 years, with the whole world being affected. Global warming is the name given to the gradual temperature rise over all the Earth. The change in temperature is small, perhaps only 2° to 3°, but it will be enough to cause large amounts of polar ice to melt. This will, in turn, raise the sea level.

According to recent reports to the British Association for the Advancement of Science, species of butterflies and birds are already moving north, where it is cooler. This is one

example of animals responding to small changes in temperature.

Such changes are also occurring in Australia.



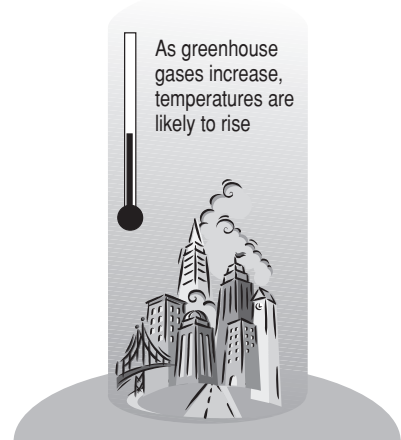
Temperature change also changes weather patterns, which could mean more rain over desert areas or less rain over farming regions. The major 2002 - 2003 drought in Eastern Australia was thought to have resulted from global warming changes. These changes could lead to wholesale extinction of different types of animals or plants and major changes to our way of life.

- 1** a What is one change due to global warming that has been noticed already?
- b Over the next few years, what are the expected temperature changes due to global warming?
- c Describe two major expected effects of this change.

The map shows Australia's South Pacific neighbours. Many of these nations are low lying coral atolls and a small rise in sea level will drown the people and the land. For example, Kiribati consists of about 30 low-lying coral islands with a total land area of over 800 square kilometres and Tuvalu, the world's second smallest country, consists of only nine small atolls.

- 2** a How could global warming changes affect you?
- b Imagine you are a citizen of Tuvalu or Kiribati. Describe your feelings about the change in sea level.

### What Causes Global Warming?



When sunlight reaches the ground and oceans after passing through the atmosphere, it heats up the land and water. During the night, this heat slowly escapes into space from the surface of the Earth. Clouds and gases in the atmosphere insulate the Earth and slow down the rate of heat loss. The heat cannot escape and the temperature gradually increases, just like in a greenhouse built to grow tropical plants. The result is called the greenhouse effect and is necessary to keep a constant environment on Earth.

- 3** a Draw (if necessary) and label the diagram above with the words: earth, sun, sunlight, heat, atmospheric gases.
- b Describe what is happening in the diagram.
- c What gases produce the greatest greenhouse change?

Greenhouse gases include carbon dioxide, methane, nitrous oxide, ozone, chlorofluorocarbons, hydrofluorocarbons and perfluorinated carbons. Carbon dioxide is believed to be responsible for 70% of global warming.

- 4** a How are these gases released into the atmosphere?
- b List some ways their production can be reduced.

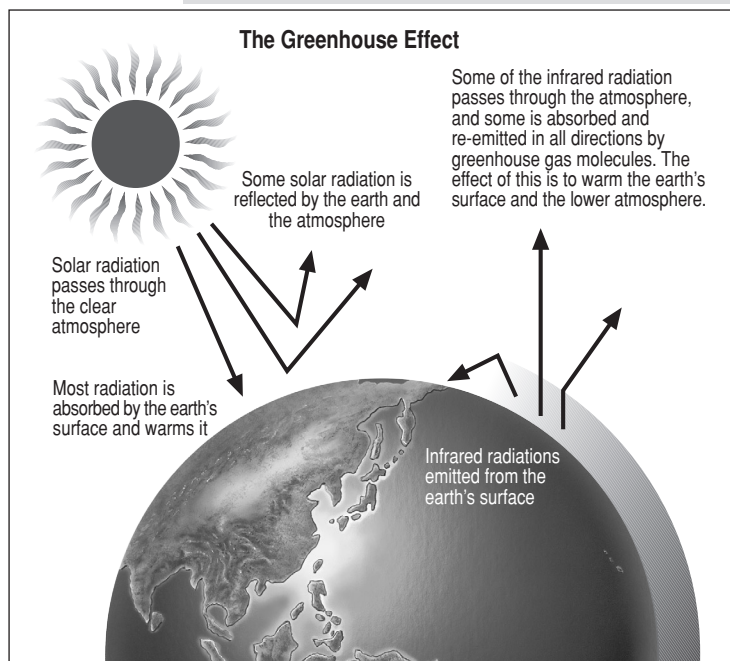
# Australia's environment

## 2i Greenhouse Effect

In Australia, the temperature rarely drops below  $-5^{\circ}\text{C}$  or rises above  $40^{\circ}\text{C}$ . If the insulating properties of the atmosphere are increased, then as the same amount of solar radiation comes in, less heat energy will be able to escape during the night. More will be reflected back to Earth or absorbed by the water vapour ( $\text{H}_2\text{O}$ ), carbon dioxide ( $\text{CO}_2$ ) and other gases in the atmosphere. This increasing level of the greenhouse effect is what is causing concern.

- 1 a What substances in the atmosphere absorb the heat?
- b Does the greenhouse effect have any advantages? If so, what are they?
- c What problems could be caused by too much carbon dioxide in the atmosphere?

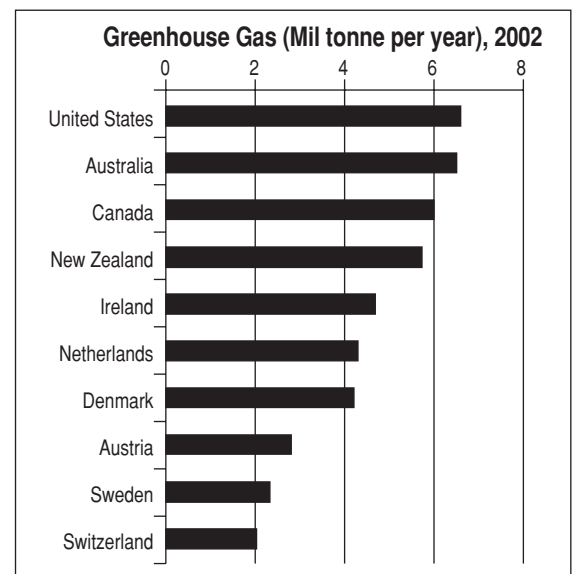
Particularly over the last decade, increases in population, technology use and dramatic increases in energy consumption have all resulted in enormous quantities of greenhouse gases being released into the atmosphere. The major problem is carbon dioxide. Before the industrial revolution, the percentage of  $\text{CO}_2$  in the atmosphere was estimated to be about 0.027%. Largely due to the burning of fossil fuels for energy and the large-scale destruction of large areas of forests for fuel and agriculture, the concentration of  $\text{CO}_2$  has risen to 0.035%. It is increasing steadily every year. The diagram below shows the process of global warming.



- 2 a What does  $\text{CO}_2$  stand for?
- b From what do we get  $\text{CO}_2$ ?
- c What is the current percentage of  $\text{CO}_2$  in the atmosphere?
- d What was the percentage of  $\text{CO}_2$  before the industrial revolution?
- e Explain what happened in the industrial revolution to cause this change.

Carbon dioxide can be formed as a product of almost any fuel reacting with oxygen. One of the major sources is car engines. The burning of petrol releases carbon dioxide as well as other pollutants.

The graph shows the greenhouse gas emissions for a number of developed countries in 2002. The information on the x-axis (horizontal) shows millions of tonnes of greenhouse gas produced per capita (person).



- 3 a Which country produces the most greenhouse gas per year?
  - b How much does a person in that country produce?
  - c How much gas do you (as the average Australian) produce per year?
  - d Suggest reasons why Sweden and Switzerland (both wealthy industrialised nations) can produce less than 30% of the gas that we produce?
- 4 Brainstorm ways that you, your friends, family and community can reduce the amount of greenhouse gas released into the atmosphere.

# Australia's environment

## 2j Research Project - Kyoto Protocol

This topic has related to the environment including pollution and global warming. To strive to reduce the effect of global warming in the future, the Kyoto Protocol or agreement was established and has been in the news and political discussions since its development. The full text of the protocol can be obtained from the first website listed below. The objective of the agreement is to: 'achieve ... stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system'.

### Kyoto Protocol

<http://www.iea.org/iea/kyoto/index.htm>

### CSIRO Australia - Scientific and Industrial Research

<http://www.csiro.au/>

### Environment Australia Home Page

<http://www.erin.gov.au/index.html>

### EPA Global Warming Site

<http://www.epa.gov/globalwarming/index.html>

### Globalwarming.net

<http://www.globalwarming.net/gwnew/default.asp>

### Global Warming - The Cooler Heads Coalition

<http://www.globalwarming.org/>

### The Natural Step

<http://www.naturalstep.org/index.html>

### Dr David Kemp, Minister for the Environment and Heritage

<http://www.erin.gov.au/minister/env/>

### US Department of State - Home Page

<http://state.gov/>

### United States Environmental Protection Agency - Topics Navigation Page

<http://www.epa.gov/epahome/topics.html>

### Weather & global monitoring

<http://www.csu.edu.au/weather.html>

### WWF - Climate Change Campaign

<http://www.panda.org/climate/>

### ABC News - Australia on its own at Kyoto climate summit <http://www.abc.net.au/news/features/kyoto/>

You are a researcher working for the office of the Minister for the Environment. You have been asked to research both sides of the 'Greenhouse Gas' debate. There are those who believe that global warming is simply within the normal climatic changes. Others say that the Kyoto protocol is too restrictive (which is why US President George W Bush has spoken out with against supporting it).

Regarding the ... Kyoto Protocol, ... the United States must emphasize that our not blocking consensus on the adoption of these Kyoto Protocol rules does not change our view that the Protocol is not sound policy. Among other things, the emissions targets are not scientifically based or environmentally effective, given the global nature of greenhouse gas emissions and the Protocol's exclusion of developing countries from its emissions limitation requirements and its failure to address black soot and tropospheric ozone. The ... treaty is not workable for the United States.

Paula J. Dobriansky, Under Secretary of State for Global Affairs, 23 July 2001  
<http://www.state.gov/g/oes/climate/index.cfm?docid=4191>

**Task** To determine the best advice for the Minister for the Environment and, ultimately, the Prime Minister, regarding the response to the Kyoto Protocol.

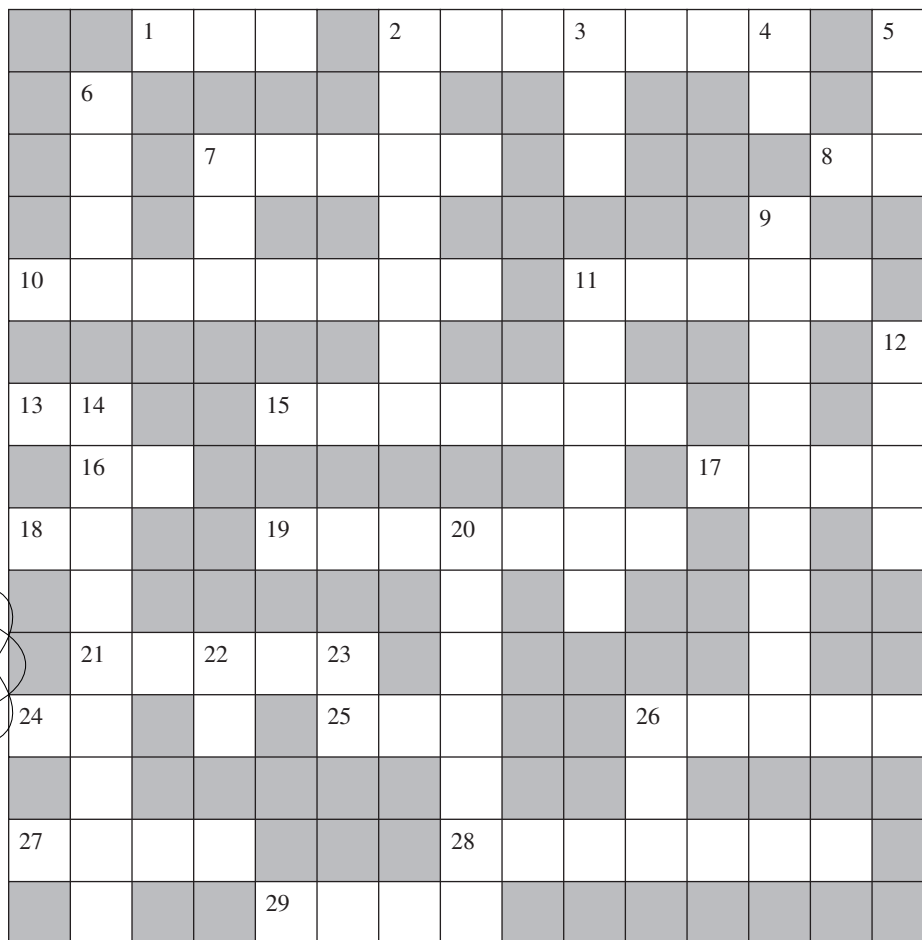
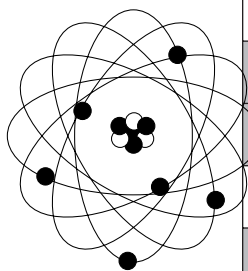
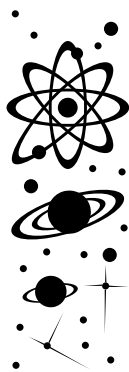
Research what the protocol says and follow up the responses for as many positions as possible. Your group report will present a unanimous response and will be supported by correctly referenced resources. Your report may include audiovisual aids, such as a computer slide show, verbal presentation and/or poster displays. You may support any response or recommendation or develop your own response.

**Current situation** Many of the websites listed above can give you current situations regarding global warming.

**Debate** If different groups in your class present alternative conclusions, present them in a debate to obtain levels of support within your class community.

# Radiation and safety

## 3a Radiation Crossword



### WORDS

alpha  
 atom  
 beta  
 Bohr  
 CAT  
 $D_2O$   
 deuterium  
 $E=mc^2$   
 EMR  
 fission  
 gamma  
 gun  
 heat  
 hydrogen  
 low  
 neutron  
 nuclear  
 nucleus  
 power  
 proton  
 radiation  
 sun  
 tritium  
 uranium  
 xray  
 Ac, Am, At, Lr,  
 Cd, Ra, Rn, Pd

### CLUES

#### Across

- 1 Chemical formula for heavy water (3)
- 2 Term for energy obtained from the nucleus (7)
- 7 Einstein's famous energy equation (5)
- 8 Symbol for the radioactive element radon (2)
- 10 Element with the smallest atom (8)
- 11 Some countries use atomic energy for the production of \_\_\_\_\_ (5)
- 13 Symbol for the radioactive element lawrencium (2)
- 15 Breaking of uranium nucleus to release energy (6)
- 16 Symbol for the radioactive element americium (2)
- 17 One product of nuclear reactions (4)
- 18 Symbol for the element cadmium (2)
- 19 Neutral particle found in the nucleus (7)
- 21 Positive particle released during fission (5)
- 24 Symbol for the element astatine (2)
- 25 Nuclear research has helped in the development of medical processes such as \_\_\_\_\_ scans (3)
- 26 EMR rays released from nuclear reactions (5)
- 27 Niels \_\_\_\_\_ was a pioneering researcher (4)
- 28 Element used in nuclear power stations (7)
- 29 Basic unit of matter (4)

#### Down

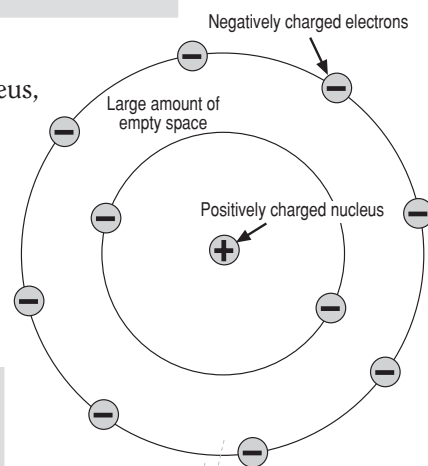
- 2 Central part of the atom (7)
- 3 This level of waste is sometimes stored underground (3)
- 4 Symbol for the element radium (2)
- 5 Closest star releases energy from fusion (3)
- 6 Radiation also used for medical imaging (4)
- 7 Electromagnetic radiation – initials (3)
- 9 Hydrogen isotope with 1 proton and 1 neutron (9)
- 11 Positive particle in nucleus (6)
- 12 Negative particle released from fission reactions (4)
- 14 Inclusive term for rays and particles released from nuclear reactions (9)
- 20 Hydrogen isotope with 1 proton and 2 neutrons (7)
- 22 Symbol for the element palladium (2)
- 23 Symbol for the element actinium (2)
- 26 Nuclear reactions can be started by firing nuclear bullets from a nuclear \_\_\_\_\_ (3)

# Radiation and safety

## 3b What is Radiation?

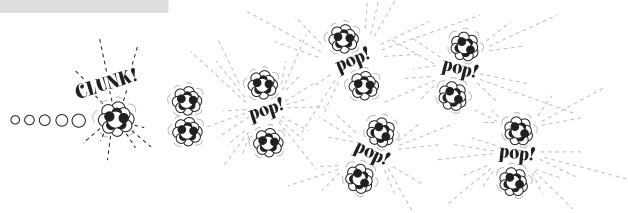
Nuclear energy—its raw material uranium, wastes and useful products, such as radioisotopes—lies on the cutting edge of science. Here risk and danger go hand in hand with usefulness and excitement about the future. Nuclear energy has caused death and sickness in many parts of the world and it could continue to cause problems if strict safeguards are not followed. The supporters of nuclear energy, however, claim that nuclear fuel provides an almost unlimited supply of clean, efficient energy.

- 1 What do you know about atoms and nuclear energy?
  - a Make up a concept map using the following words: nucleus, proton, neutron, radiation, energy, fuel, uranium.
  - b How is Australia involved in the nuclear industry?
- 2 The diagram shows a model or picture of an atom.
  - a What charge do electrons have?
  - b Protons are positively charged. Where are they found?
  - c What makes up most of the volume of an atom?



Nuclear energy is usually obtained from the large nucleus of one atom (usually uranium) breaking into small parts. This releases a lot of energy.

- 3
  - a List two advantages of nuclear power.
  - b Explain how one of these advantages might affect you.
  - c List two disadvantages of nuclear energy.
  - d Explain how one of these disadvantages might affect you.



The diagram above shows fission of a uranium nucleus after being hit by a neutron. Fission is the breaking up of a large molecule. Some lines represent particles or energy given off but not colliding with another atom.

- 4 What happens to the particles or energy which does not hit another uranium nucleus?

A nuclear reaction is only one source of radiation. The sun releases continuous solar radiation and we are urged to protect our skin and eyes from this radiation. We are also bombarded with electromagnetic radiation from a whole range of sources. These include radio transmitters, microwave ovens, mobile telephones and other communication material.

As part of the future you will be expected to make decisions about the use of nuclear energy and other nuclear products as well as the many types of electromagnetic radiation produced in our world. You must weigh up the possible benefits to our lifestyle and society and compare these to the concerns and problems.

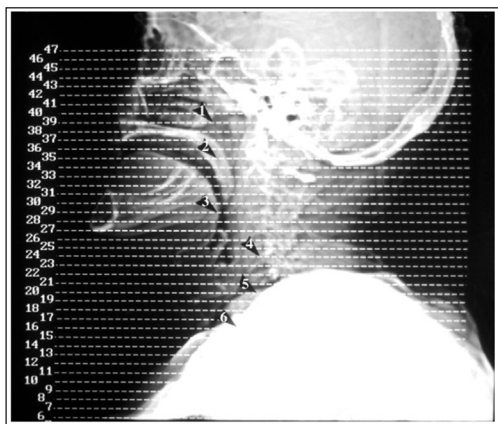
- 5
  - a Should we be concerned about solar radiation?
  - b What damage can too much solar energy do to our skin or eyes?
  - c What do you do to protect yourself from solar radiation?
  - d Is there a difference between radiation from an atomic reaction and electromagnetic radiation from the sun or radio/telecommunications equipment? Do you think the difference is important?

# Radiation and safety

## 3c Radiation and Health

Whenever strong radiation affects living tissue the first effect is burning. The molecules in the cell cannot absorb the energy from strong radiation. Many thousands of chemical bonds are destroyed and the protein material in the cells and tissues is cooked. This has been most noticeable in the effects of atomic explosions in Japan and in radiation accidents in the former USSR and Japan.

It is generally accepted that low level radiation, such as diagnostic x-rays (when correctly used by radiographers to check bones and teeth) do not cause burning. This means that the exposure to the radiation is less than that required to physically destroy the body's cells. It does not mean that the radiation is perfectly safe.



**1** The graphic shows an x-ray of a head and neck.

- a How do x-rays help doctors?
- b Suggest reasons why the person photographed here may have had an x-ray of the neck and head.
- c List some safety requirements for taking x-rays.
- d Should you worry if you had to have one x-ray for a broken toe?
- e What about a set of x-rays for a broken pelvis? Explain your concerns if your answer changes.

Radiographers always stand behind a screen or leave the room when an x-ray photograph is being taken.

- 2**
- a What does a radiographer do?
  - b Are x-rays more dangerous for the operators? Explain.
  - c Would x-rays be more dangerous for you if you needed a large number of x-rays? Explain why.



There are two main types of radiation: ionising and non-ionising. Ionising radiation either adds or removes electrons to the material they contact whereas non-ionising radiation does not. Ions, or charged atoms, are made by changes to the numbers of electrons. Ionising radiation, such as gamma ( $\gamma$ ) rays and x-rays, is produced from nuclear reactions, during the reaction as well as from radioactive material. This type of radiation is also found in very small amounts naturally in the environment. It is related to the ores of radioactive metals or as a part of solar energy. Non-ionising radiation is emitted by electronic equipment including power lines, telephone and radio transmitters and it is referred to as EMR or electromagnetic radiation.

- 3**
- a What does EMR stand for?
  - b What produces EMR?
  - c List some of the ways we can detect and use EMR.
- 4**
- a Signs around x-ray machines warn pregnant women of the dangers. Are x-rays dangerous to the mother or the unborn child?
  - b Describe any sort of danger they could pose.
  - c Are there other radiation dangers for pregnant women? Consider the concerns people have about high voltage power lines, microwave ovens and cellular phones and list some of these concerns.
  - d Are these concerns justified? Present arguments both for and against people being concerned.
  - e How could you find out information about the possible health problems of these items?

Further useful information about x-rays can be obtained from the interesting and interactive University of Colorado website  
<http://www.colorado.edu/physics/2000/xray/>

# Radiation and safety

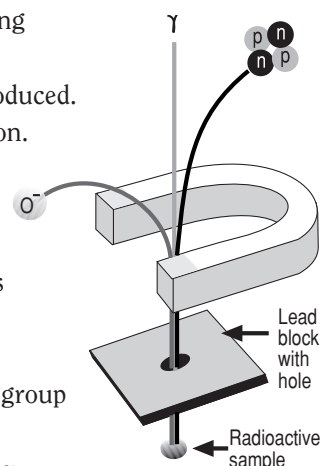
## 3d Ionising Radiation

The four main types of ionising radiation are:  $\alpha$  (alpha) particles, which are large and stopped by a few layers of skin;  $\beta$  (beta) particles, which are small and can pass into the skin and tissue;  $\gamma$  (gamma) rays; and positrons, which are positively charged electrons ( $\beta^+$ ). Gamma ( $\gamma$ ) rays are not particles at all but part of the electromagnetic spectrum. Light rays are also part of the electromagnetic spectrum but unlike light,  $\gamma$  rays are able to pass right through the body.

The four main types of radiation are described in the table.

| Radiation                  | Description                        | Mass                                                   | Charge          | Penetration in air | Penetration in the body | Electric/magnetic fields                |
|----------------------------|------------------------------------|--------------------------------------------------------|-----------------|--------------------|-------------------------|-----------------------------------------|
| $\alpha$ (alpha) particle  | 2 protons, 2 neutrons (He nucleus) | $6.7 \times 10^{-27}$ kg (7363 $\times$ mass electron) | +2              | few centimetres    | few millimetres         | deflected                               |
| $\beta$ (beta) particle    | electron                           | $9.1 \times 10^{-31}$ kg                               | -1              | few metres         | few centimetres         | deflected opposite to $\alpha$ particle |
| $\beta^+$ (beta+) particle | positron (positive electron)       | $9.1 \times 10^{-31}$ kg                               | +1              | few metres         | few centimetres         | deflected same as $\alpha$ particle     |
| $\gamma$ (gamma rays)      | electromagnetic radiation - EMR    | 0 kg (zero mass)                                       | 0 (zero charge) | many metres        | whole body              | no deflection                           |

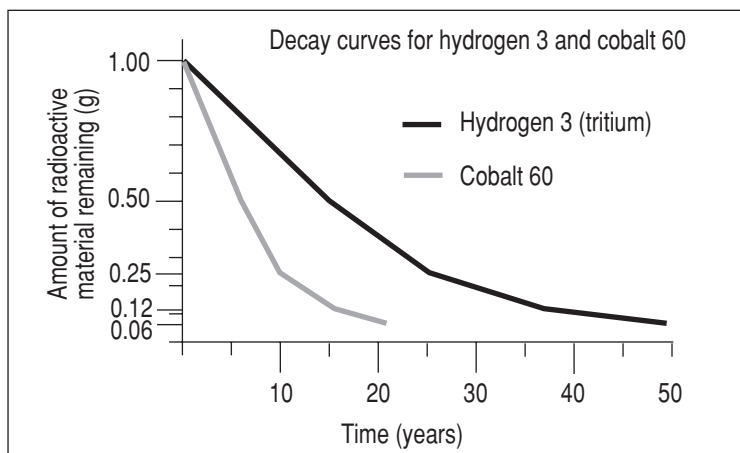
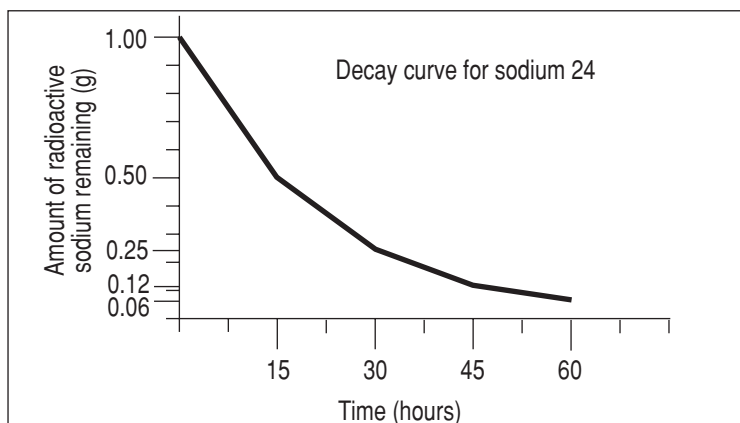
- 1
  - a What makes up an  $\alpha$  particle?
  - b Give another name for a  $\beta$  particle.
  - c What is the charge of a positron?
  - d How far can  $\gamma$  rays travel in air?
  - e Why do  $\gamma$  rays have no mass?
- 2
  - a What do you understand by the word 'penetration' as used in the table?
  - b Explain the different penetration properties of the particles and rays.
- 3
  - a List the main differences between the different types of ionising radiation.
  - b State some possible ways these types of radiation could be produced.
  - c Compare the mass of an  $\alpha$  particle with the mass of an electron. How much larger is the  $\alpha$  particle?
  - d Use this difference to describe how the  $\alpha$  particle affects the cells in the body.
- 4
  - a Why do the particles deflect in an electric or magnetic field as shown in the diagram?
  - b Suggest reasons why they deflect in different directions.
  - c Which deflects the most? Which deflects the least? In a small group explain why.
- 5
  - a If someone is exposed to radiation, what part of the body will  $\alpha$  particles affect most?
  - b Do you think radiation might cause microscopic changes to the body that we cannot see? If so, what parts of the body could be most affected? Discuss this concept with your group. The Berkeley Laboratory at the University of California has lots of information on its site <http://pdg.web.cern.ch/pdg/particleadventure/frameless/index.html>.



# Radiation and safety

## 3e Continuing Nuclear Radiation

A part from the radiation which is released during the process of fission, the waste materials from nuclear reactions are themselves radioactive. These substances release alpha, beta and gamma radiation for long periods of time after they are made. The graphs below show the decay or breakdown curves for three radioactive substances: sodium, hydrogen and cobalt. In each case, the number refers to the number of protons and neutrons in the nucleus of each isotope (or type of atom). The particular isotope of sodium has 24 protons and neutrons, hydrogen has 3 and cobalt has 60.



The radioactive isotope sodium 24 emits or gives off radiation and changes into a different element. If we have one gram (1 g) of Na 24 then after 15 hours we will have only 0.5 g left. This period of time is called the half-life. Releasing the  $\alpha$ ,  $\beta$  or  $\gamma$  rays has changed the rest of the sodium into something else.

- 1 a How many hours before this amount is itself halved?
- b The next half-life takes how long?

- c What is the half-life of the sodium 24 isotope?
- d Define 'half-life' in your own words.
- 2 a What is happening during the 60 hours shown on the first graph?
- b What happens after the 60 hours are up?
- c When does the sodium 24 stop emitting radiation?
- 3 a Use the second graph to determine the half-lives of hydrogen 3 and cobalt 60.
- b If 100 tonnes of cobalt 60 is released into the atmosphere after an accident, how much is still present after 20 years?

The table shows the half-lives of a number of different radioactive materials and the radiation that is emitted from each.

- 4 a Determine the half-life of strontium 90.
- b What type of radiation is emitted from the decay of carbon 14?
- c If 32 g of Fluorine 20 is released in a laboratory, calculate how long before there is only 1 g left?

Plutonium is a major waste produced in nuclear reactors.

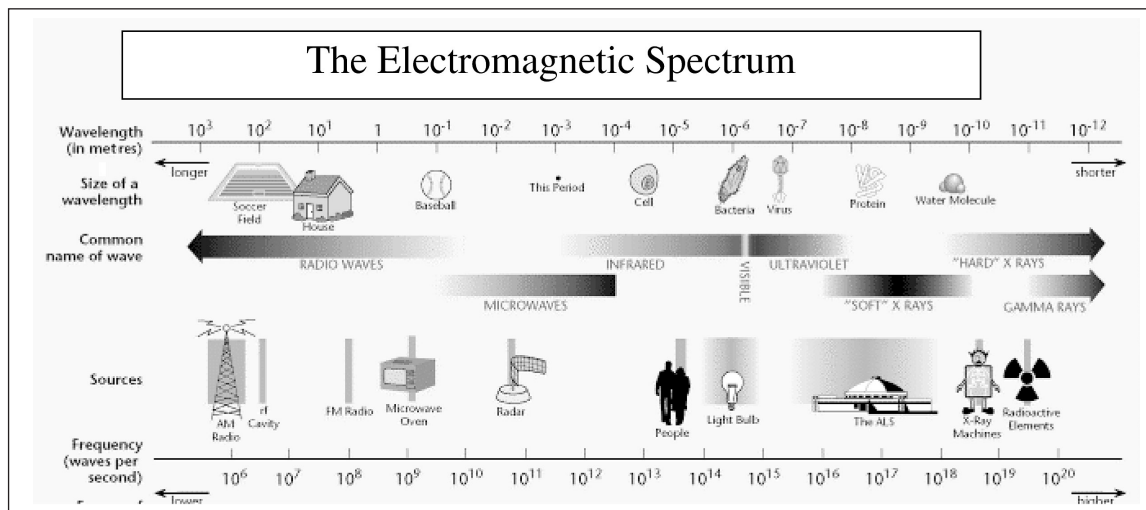
- 5 a How long will 25 tonnes of Plutonium 239 have to be stored?
- b Discuss in a group ways of reducing the radiation from nuclear waste material.
- c Graph the decay curve of one of the isotopes from the table.

| Isotope name  | Radiation           | Half-life       |
|---------------|---------------------|-----------------|
| Fluorine 20   | $\beta$ , $\gamma$  | 11 seconds      |
| Magnesium 27  | $\beta$ , $\gamma$  | 9.5 minutes     |
| Sodium 24     | $\beta$ , $\gamma$  | 15 hours        |
| Iodine 131    | $\beta$ , $\gamma$  | 8 days          |
| Cobalt 60     | $\beta$ , $\gamma$  | 5.3 years       |
| Hydrogen 3    | $\beta$             | 12.3 years      |
| Strontium 90  | $\beta$             | 28 years        |
| Radium 226    | $\alpha$ , $\gamma$ | 1600 years      |
| Carbon 14     | $\beta$             | 5700 years      |
| Plutonium 239 | $\alpha$ , $\gamma$ | 24 000 years    |
| Uranium 238   | $\alpha$ , $\gamma$ | 4500 mill years |

# Radiation and safety

## 3f Non-ionising Radiation

**N**on-ionising radiation is found in the form of the energy waves in the electromagnetic spectrum. This spectrum is a label given to a large range of radiations each of which involves wave like (cyclical) changes to an electric charge together with corresponding changes to the strength of a magnetic field.



Like any wave, electromagnetic waves have a specific frequency and wavelength. The wavelength (referred to as  $\lambda$  and measured in metres) is the distance between a particular part of a wave and the same part on the next wave. The frequency of the wave is defined as the number of whole wavelengths that pass a particular point per second. Frequency is measured in cycles/second or Hertz (Hz). All waves have the same relationship between wavelength, frequency and velocity. That is velocity equals frequency multiplied by wavelength. In a formula this is written as  $v = f\lambda$ . The energy of these waves is related to their frequency so the higher the frequency, the higher the energy.

For all electromagnetic radiation the velocity is the same as the velocity of light ( $3.0 \times 10^8 \text{ ms}^{-1}$ ) and only varies when travelling through different substances, such as glass or plastic. In fact, visible light forms only a small part of the large set of waves in the electromagnetic spectrum. The difference, therefore, between the types of electromagnetic radiation is the frequency and corresponding wavelength. Some radio waves have wavelengths of over a kilometre and frequencies of around 100 million hertz (100 MHz).

- 1**
  - a A soccer field is  $10^2$  metres long. How long is that?
  - b What EMR wavelengths have the same length (or longer) than a soccer field?
  - c Approximately how long are the EMR waves which we detect as visible light?
  - d How can we detect: radio waves, infrared waves, light waves and x-rays?
  - e What is the frequency (hertz or cycles per second) of: microwaves, light, radar and gamma radiation?
  - f Which waves have the greatest energy?
- 2**
  - a Using the formula above calculate the wavelength of ABC Classic FM in Sydney (92.9 MHz).
  - b Calculate the wavelength of a popular AM radio station.
  - c Calculate the wavelength of your favourite FM radio station.
  - d Microwaves have a wavelength of about 3 cm. What is their frequency?

# Radiation and safety

## 3g Safety and Non-ionising Radiation

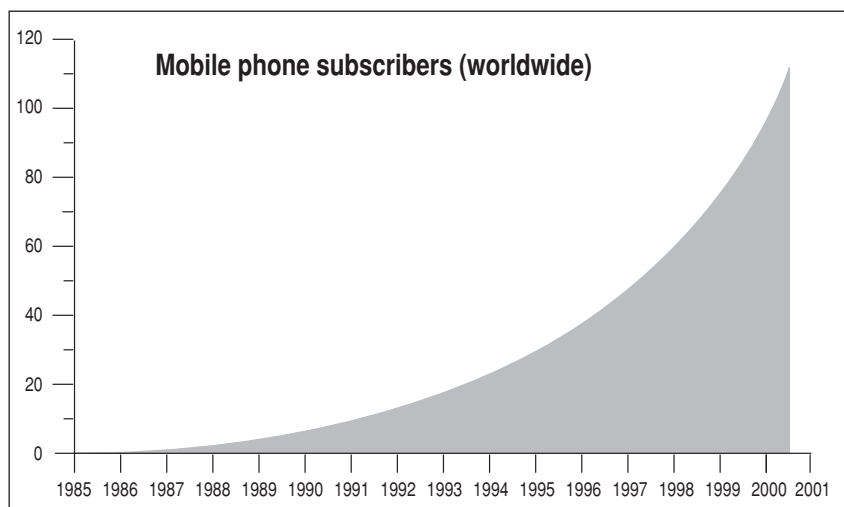
Radio waves are used by telephone systems to cover remote areas of Australia. Telstra uses solar-powered repeater stations about 50 km apart. Radio waves can be bounced off layers in the Earth's atmosphere and will travel around hills and buildings. That is why you can listen to the radio even inside the house.

As shown in the graph, mobile phone use continues to rise. This is why scientists and health professionals are so concerned about potential risks associated with their use.

Mobile phones are just one part of the radiation increasing in our environment every day.

There are many fears regarding electromagnetic radiation. One publicised example includes children living near high-voltage power lines having a higher proportion of leukaemia (type of cancer) than the general population. Cancer can be caused by chemicals (such as in cigarette smoke and pesticides), ultra-violet radiation from the sun and radiation from atomic reactions or isotopes in the earth or the atmosphere.

As well as affecting the adult cells, radiation may change the genetic code. A change in a developing sex cell will produce a mutation in the genes and chromosomes that make up the cell. Depending on the position on the chromosome and the amount of change, it may not be observed in individuals for a number of generations.



Mobile Phone Use, 2000

| Country     | %  |
|-------------|----|
| Hong Kong   | 75 |
| Norway      | 69 |
| Taiwan      | 69 |
| Sweden      | 68 |
| New Zealand | 68 |
| South Korea | 66 |
| Italy       | 65 |
| Australia   | 64 |
| Ireland     | 60 |
| UK          | 60 |
| Singapore   | 59 |
| USA         | 53 |
| Germany     | 51 |
| France      | 46 |

- 1 a How many mobile phone users were there in 1992 and 1995.  
b How many extra subscribers were there between these years?  
c From 2000, how many years will it take to add another 20 million subscribers?

The table shows the percentage of people (over 16 years) who own mobile phones.

- 2 a Graph this information in a bar graph.  
b Suggest reasons why Australia's mobile phone usage is higher than USA and lower than Hong Kong.  
c What predictions could you make about countries such as China and India that are not in the above 46% category yet?

- 3 a When cells divide (undergo mitosis or meiosis) there is a greater possibility of cancer. Why is this so?  
b What happens during cell division which makes cells more susceptible to radiation?  
c What advice could you give someone who wants to reduce the possibility of getting cancer?

- 4 a What is the difference between different types of electromagnetic radiation? Is this difference important?  
b We use televisions, radios, microwaves and other electrical items everyday. Should we stop using these items because of our fears? Discuss individual responses in a group.  
c Is it safe as long as we are not close to large and powerful sources? Is there a 'safe' level?

- 5 a Is it possible to 'see' or 'feel' electromagnetic radiation?  
b If we are unable to detect it with our bodies, describe the technology available for detecting EMR.

# Radiation and safety

## 3h Extra-low Frequency

The electromagnetic radiation at the opposite end of the spectrum to gamma rays and x-rays was considered for many years to be harmless to people and useful to society. These waves include radio and television waves as well as microwaves. Only recently is there increasing concern about how these extra-low frequency (ELF) waves affect the cells in our bodies.



High-voltage cables which cross the countryside have voltages around 500 000 volts (500 kV). A current flowing with this voltage produces an extremely powerful magnetic and electric field

around the wires. People living and working in close proximity to the cables are immersed in these fields. The fields are so strong that they can cause fluorescent bulbs to glow, hair to stand on end and produce charges on cars parked under the wires. There is much concern in the community as to what effect these magnetic and electric fields have on living tissue, particularly that of growing children. There have been a number of studies done on children in areas affected by such high-voltage cables, but even though these results point to increases in cancers and birth defects the studies can only suggest reasons to explain these changes. The results cannot prove that the high-voltage fields caused cellular destruction.

- 1**
  - a What waves are found at the two ends of the electromagnetic spectrum?
  - b List some of the properties of each of these waves.
  - c What do the initials ELF refer to?
  - d What is a common voltage found in high-voltage powerlines?
  - e What voltage do you use in your house?
- 2**
  - a What are produced around wires which carry high voltages?
  - b Describe some of the effects of these fields.
  - c List some of the concerns of residents about high voltage wires.

Many people use household items that produce magnetic and electric fields. Items causing the most concern are electric blankets, hair dryers, mobile phones and computer/TV screens. Even toasters produce small electric and magnetic fields but toasters are only used for a few minutes. Electric blankets sometimes bathe the sleeping person in electromagnetic fields for hours at a time. Hair dryers and mobile phones also produce the fields very close to the brain. Televisions and computers are a concern particularly because young people use them for long periods of time.

- 3** Continue the table started here on your own paper. The table should show some of the items that you might use which release low frequency waves. Any household item with an electron gun (such as a TV or computer) or an electric motor (including the fridge) should be included. Also include any item that sends out a signal, such as mobiles or cordless phones. An example is done for you.

| Household item | Radiation producer | Used by (name of person) | Use per day (hours) |
|----------------|--------------------|--------------------------|---------------------|
| computer       | monitor/screen     | myself                   | 4                   |
|                |                    |                          |                     |

- 4**
- a What is the frequency (in Hz) for our household electricity?
  - b How many television or radio stations can you receive at your house?
  - c Are all the radio and TV waves travelling through your house all the time?
- 5**
- a Do you use any of the following items: electric blanket, computer, television, mobile phone, toaster, microwave oven? Discuss your use in a small group.
  - b Would you modify your use in the light of concerns by scientists about electromagnetic radiation?
  - c Would you want to live near a high-voltage power line? Discuss your reasons with your group.
  - d List some of your concerns.

# Radiation and safety

## 3i Health Concerns and the Community

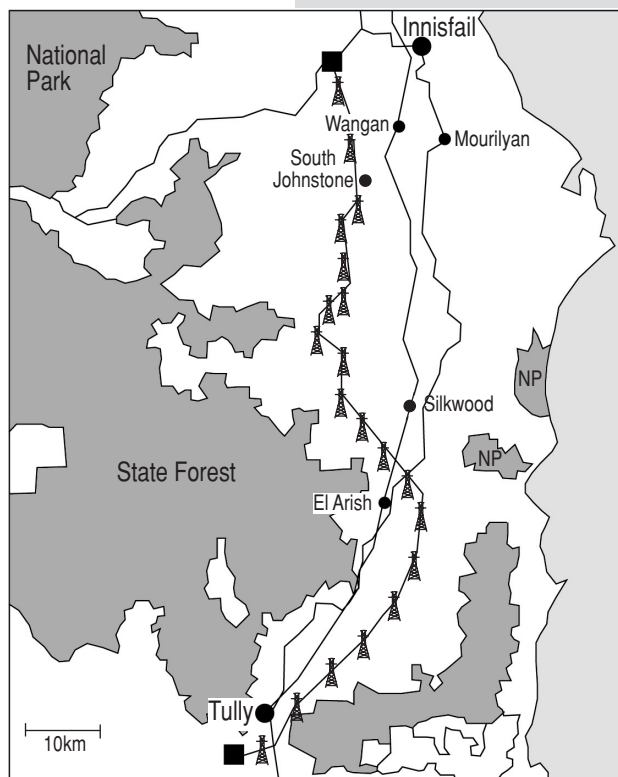
### CASE STUDY - NORTH QUEENSLAND

Many people living in northern Queensland are concerned about a proposal to build high-voltage powerline near their towns and farming areas. The proposed 275 kV line (to replace the current 137 kV line) is expected to go from near Tully to Innisfail in north Queensland. An environmental impact statement (EIS) was completed and posted on Powerlink's website in early 2004. (Powerlink Queensland is a government-owned corporation that owns, develops, operates and maintains the state's high-voltage electricity transmission network.)

Powerlink maintains that increased power is a necessary part of society because we all use electricity in our day-to-day life and there are increasing demands for it. Residents in Tully and the local Shire Council are concerned and have criticised Powerlink for failing to address their apprehension.

There are concerns because the powerline route is within 200 metres of Tully State High School land and crosses the approach to the Tully airfields, adversely affecting aerial spraying and the Tully skydiving industry. It will also increase costs to cane growers. The powerline will result in a drop in property values and have a great

visual impact as well as compromising the Walter Hill Range wildlife corridor, which could affect tourism. People also believe it will be harmful to their health.



- 1**
  - a** On the map of Australia, mark in the Tully/Innisfail area of North Queensland. Also mark where you live.
  - b** Research other places where there are high-voltage wires and mark these. Make an observation about their location.
- 2**
  - a** What are some of the concerns that the residents of Tully have about the proposed upgrade to their powerline? Rank these concerns in order that you think are most important to least important.
  - b** Write a letter to your local Federal or State Member of Parliament, putting forward your views on the dangers or advantages of high-voltage powerlines.

The Electromagnetic Radiation Alliance of Australia website <http://ssec.org.au/emraa/> looks at the health issues of a large number of electrical items including computers, mobile phones and even hair dryers.

- 3**
  - a** In forming your opinion on this and other issues, should you only accept the information in this book or should you research more widely?
  - b** The Electromagnetic Radiation Alliance site presents a lot of information on health and radiation. Is it useful as a sole source of information?
  - c** In a group, describe the process you would carry out to gain as much information as possible on this topic so you can make informed decisions.



# Radiation and safety

## 3j Research Project: Radiation Effect on Human Life

This topic has related to the radiation released from both nuclear and electronic sources and its affect on our wellbeing. The concerns people have about radiation extend beyond the fear of radioactive fallout to the electromagnetic radiation produced by our day-to-day computers, phones and electric motors.

In this topic a number of websites have been listed. There are many other sources of information about radiation and its dangers. Some include:

Australian Communications Authority

<http://www.sma.gov.au/>

Berkeley Campus (UC)

<http://pdg.web.cern.ch/pdg/particleadventure/frameless/index.html>

CNN Health Page

<http://www9.cnn.com/HEALTH/9803/24/cancer.environment/>

Electromagnetic Radiation Alliance of

Australia <http://ssec.org.au/emraa/>

Institute for Communications Sciences

<http://www.its.bldrdoc.gov/>

Medical College of Wisconsin

<http://www.mcw.edu/gcrc/cop.html>

Nasa Education

<http://observe.ivv.nasa.gov/nasa/education/reference/main.html>

Parents Against Leukaemia

<http://www.pal-uk.org/>

Radiation Oncology

<http://www.cancer.com.au/index.html>

University of Colorado

<http://www.colorado.edu/physics/2000/index.pl>

University of London Department of

Computing

<http://www.doc.ic.ac.uk/>

Uranium Information Centre

<http://www.uic.com.au/>

US Nuclear Regulatory Commission

<http://www.nrc.gov/nrc.html>

Victoria Radiation Safety Unit

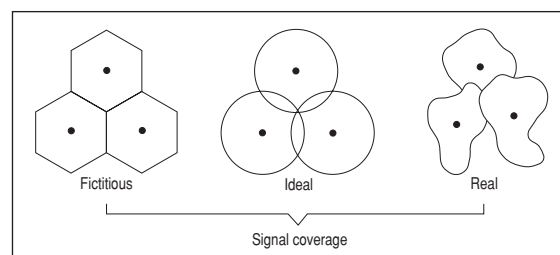
<http://www.dhs.vic.gov.au/phd/hprot/rsu/>

Imagine you are in a partnership of barristers arguing in court for either cancer victims or power companies. Your client needs you to fight for compensation and a removal of high-voltage powerlines OR for the powerlines to stay without further claim. Your client is Australian but most research has been done overseas in the US and Europe and you will have to search for the results of studies to support your case.

**Task** To determine the best response for your client after carrying out research on the topic. Your group report will present the unanimous response and will be supported by correctly referenced resources. Your report may include audiovisual aids, such as a computer slide show, verbal presentation and/or poster display. You should develop your own response.

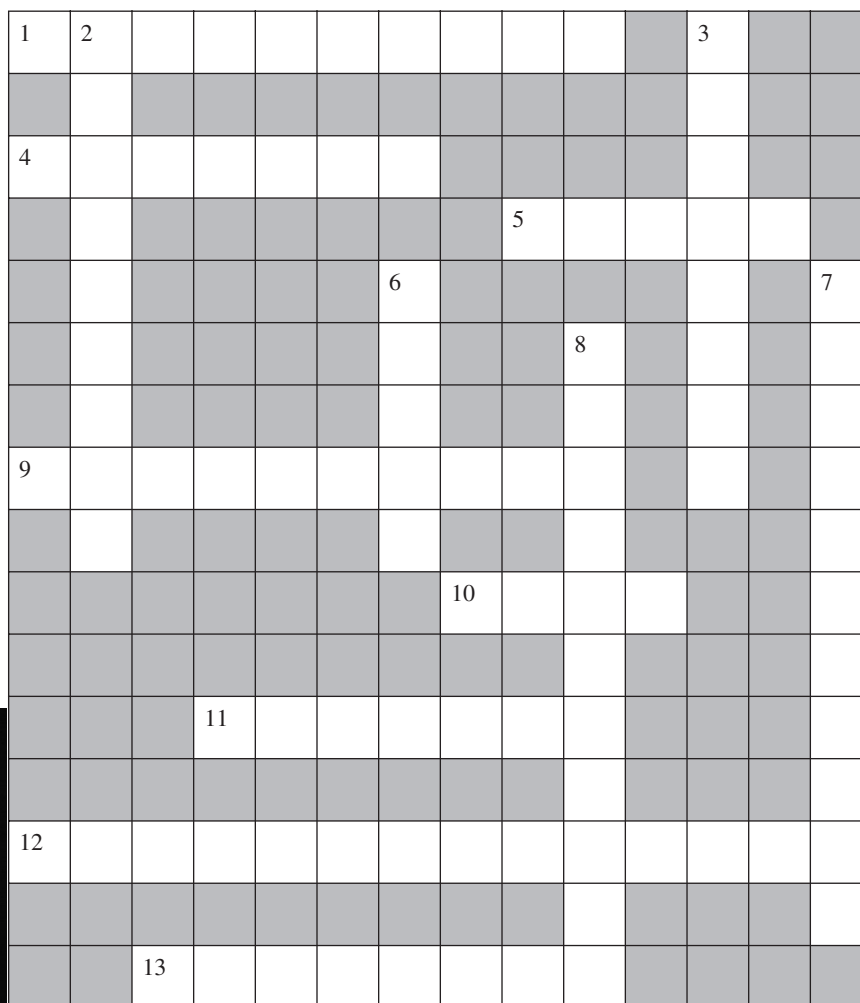
**Debate** If different groups in your class present alternative conclusions, present them in a debate to obtain levels of support within your class community. Note: the websites listed above, and others you will find, present one point of view. Your task will be difficult because you will need to compare the different claims from the different sides in the debate.

**Question** The diagram shows the organisation of mobile phone cells (which is why they are called 'cell phones' in the USA). The fact that these cells overlap means that we are in the electromagnetic radiation of not only our own but also other people's mobiles. Should this concern us? What do you think?



# Everyday chemistry

## 4a Chemistry Crossword



### WORDS

adhesion  
colloids  
emulsifier  
mayonnaise  
oil in water  
polar  
soap  
solvent  
surfactants  
surface tension  
suspensions  
tyndall  
water

### CLUES

#### Across

- 1 Used to allow for a substance to remain suspended in a solution for a greater amount of time. It is often used in ice-cream to leave it smooth and consistent. (10)
- 4 Name given to the process or way of testing whether a substance is a solution, suspension or a colloid. (7)
- 5 Type of emulsion that gives a warming effect often used as a sun screen. It is a \_\_\_\_\_ in oil emulsion (5)
- 9 Type of emulsion that gives a cooling effect. It is often used in cold creams. (3, 2, 5)
- 10 Cleaning agent made from fat, sodium and potassium hydroxide. (4)
- 11 Liquid that dissolves a solute to form a transparent solution. (7)
- 12 Describes the strong forces between molecules in a liquid that pull the liquid inwards and forms a barrier on the surface. (7, 7)

- 13 Mixture that remains suspended for a long period of time, for example paint. (8)

#### Down

- 2 An example of a colloid. (10)
- 3 The term used to describe the attraction of the water climbing or being attracted to glass. (8)
- 6 Molecule that has a positive and a negative end. (5)
- 7 Mixtures that can form layers. (11)
- 8 Used to suspend fats from plates when cleaning dishes. (11)

## 4b Solutions, Suspensions and Colloids

Pure substances contain identical particles, for example gold only contains gold atoms and a jar of salt only contains salt crystals. Mixtures, however, contain two or more substances, for example a cake contains flour, sugar, eggs etc. Solutions, suspensions and colloids are all examples of mixtures.

- 1 Write descriptions of the following words:
  - atom
  - molecule
  - particle
  - salt
  - gold.
- 2 a Using the internet, research the following terms: solutions, suspensions and colloids.  
b Write a description of each.
- 3 Now that you know the difference between the different types of mixtures, complete the following table by identifying the substance and whether it is a **solution, suspension or colloid**.

| Substance      | Solution, Suspension or Colloid |
|----------------|---------------------------------|
| Salt water     |                                 |
| Muddy water    |                                 |
|                | Colloid                         |
| Mayonnaise     |                                 |
| Milk           |                                 |
|                | Solution                        |
| Soft-drink     |                                 |
| Salad dressing |                                 |
|                | Suspension                      |
| Sponge cake    |                                 |

- 4 a Choose one solution from the table and explain how we use its solution properties in our lives.  
b Pick the most useful suspension and explain why it is useful to you.  
c Is there a colloid which cannot be replaced by a suspension or solution?  
d Describe why it is so unique.
- 5 a What are some of the characteristics of paint?  
b Classify paint as a solution, suspension or colloid.  
c Explain the reasoning for your classification.
- 6 a Describe how correction fluid (Liquid Paper) is used.  
b Is this substance a solution, colloid or suspension?  
c Research correction fluid and paint to help in your decision making.

Two useful web sites are: <http://enrich.sdsc.edu/SE/cpaintstud.html> and <http://www.dhfs.state.wi.us/eh/ChemFS/fs/TCE.htm>

### INVESTIGATION - MAYONNAISE

An example of a colloid is mayonnaise. Mayonnaise can be made by following the list of instructions below. Give it a go!

1. Place 1 egg yolk in a small bowl and mix in salt, pepper and mustard to taste.
2. Measure 3 tablespoons of oil in a cup and 1 tablespoon of vinegar in another cup.
3. Add a few drops of the oil to the egg yolk in the small bowl and mix in gradually with a spoon.
4. Add a few drops of the oil to the egg yolk in the small bowl and mix gradually with a spoon. Continue slowly until all the oil and vinegar is used.
5. Now taste it!

# Everyday chemistry

## 4c Tyndall's Effect

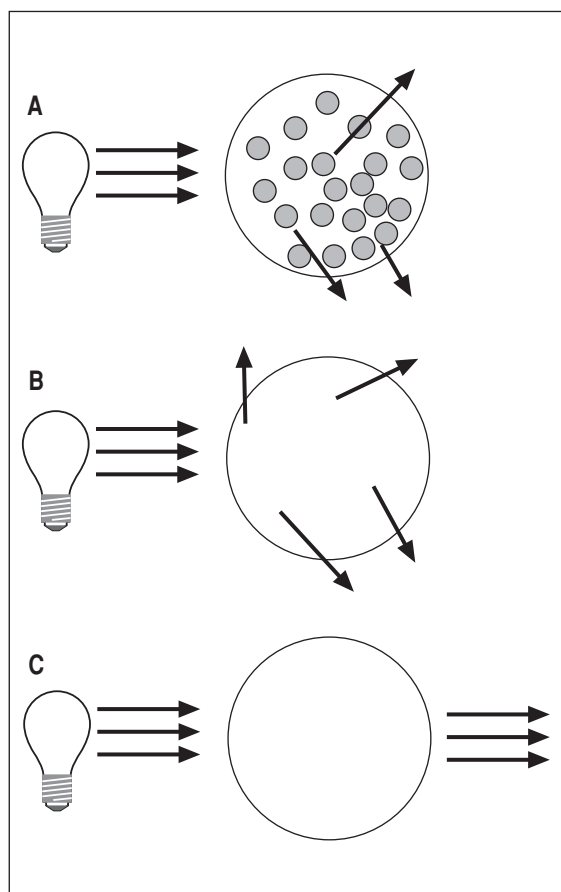
**T**yndall's effect is a method used to determine whether a substance is a solution, a suspension or a colloid. When light travels through a substance it refracts or bends. When this bending is measured, the particular qualities of the material can be compared with others. If the suspension or solution is allowed to stand for some time, a suspension will form layers. This will change the appearance of the light passing through it. Light cannot pass through a colloid because the particles are too large.

- 1** Write a procedure you could follow to determine whether a substance was a suspension, a solution or a colloid.
- 2** Research Tyndall's effect and describe how it works. The following website may assist you — you may also be able to complete the experiment yourself!  
Solutions, suspensions and colloids lab:  
<http://auden.webster.edu/~anderjoe/packets/pack8/labcolloids.pdf>
- 3** The diagrams below labelled A, B and C represent a solution, a suspension and a colloid.
  - a** Next to each write whether it represents a solution, a suspension or a colloid.
  - b** For the solution, suspension and colloid explain why the light travels the way it does when tested for Tyndall's effect.

This experiment allows us to work out the difference between solutions and suspensions or colloids.

- 4** Using the diagrams, explain why the light travels straight through in C but is refracted (bent) in examples A and B.
- 5** Explain how you could easily determine the difference between a colloid and a suspension.
- 6** Use your answer to complete the table below.

You can find out more about John Tyndall by going to [http://www.tyndall.ac.uk/general/john\\_tyndall.shtml](http://www.tyndall.ac.uk/general/john_tyndall.shtml).



| Characteristic                     | Solution   | Colloid | Suspension |
|------------------------------------|------------|---------|------------|
| Particle size                      |            |         | largest    |
| Is light refracted?                |            | yes     |            |
| Can the particles be filtered out? | no         |         |            |
| Do particles settle out over time? |            |         | yes        |
| Uniformity of the mixture          | homogenous |         |            |

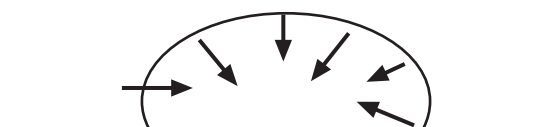
## 4d Surfactants and Surface Tension

### ACTIVITY 1

Take a five cent piece and a dropper. Fill the dropper with water and see how many drops of water you can balance on top of the coin. You may be surprised!

- 1** How many drops can you fit on a five cent coin?

The activity that you have just performed demonstrates **surface tension**. It can be explained using the diagram below.



- 2** Label the diagram with the words: air, solution, coin, attractive force

The attractive forces at the surface of the substance are greater with each other (ie in the solution) than with the different particles in the air, therefore you see the rounding of the droplet where it attaches to the table or coin.

### ACTIVITY 2

Take a small basin or petri dish and fill with water. Take a needle and balance or float it on top of the water.

- 3** What happens to the needle?

This is surface tension at work! Now that you have it floating, without rocking or knocking the dish, place a drop or two of detergent in the water and observe what happens.

- 4** What happens to the needle after detergent is added?

The detergent lowers the surface tension, it is said to be a **surfactant** or surface active agent. Oil is removed from your dirty dinner plates in the same manner.

- 5** Describe how the surfactant or detergent works on the oil that covers your plate after dinner. Use a diagram to help with your explanation. The following website may assist you with this:

[http://www.kcpc.usyd.edu.au/discovery/9.5.5-short/9.5.5\\_introsurfactants.html](http://www.kcpc.usyd.edu.au/discovery/9.5.5-short/9.5.5_introsurfactants.html)

- 6** Write definitions for the following terms: Anionic, Cationic, Non-ionic, Amphoteric, Polar, Non-polar, Hydrophilic, Hydrophobic.

- 7** a Match the following diagrams of surfactants with the correct label.  
b Label the hydrophobic and the hydrophilic ends on each diagram.

A Anionic

B Cationic

C Non-ionic

D Amphoteric

- 8** Each type of surfactant has a different function or purpose. Research this and determine the use for each type of surfactant. <http://surfactants.net> will help in your search.

### Investigation - Bread tie races

Collect a few plastic bread ties from your bread packets at home. Take a shallow baking dish or plate and fill it with water. Float the bread ties in a straight line in the dish; this will be your starting line. Now take some detergent (you may like to compare a few different brands to see which work best) and place one drop into the circular part of the bread tie. See which one travels the furthest. Good luck!

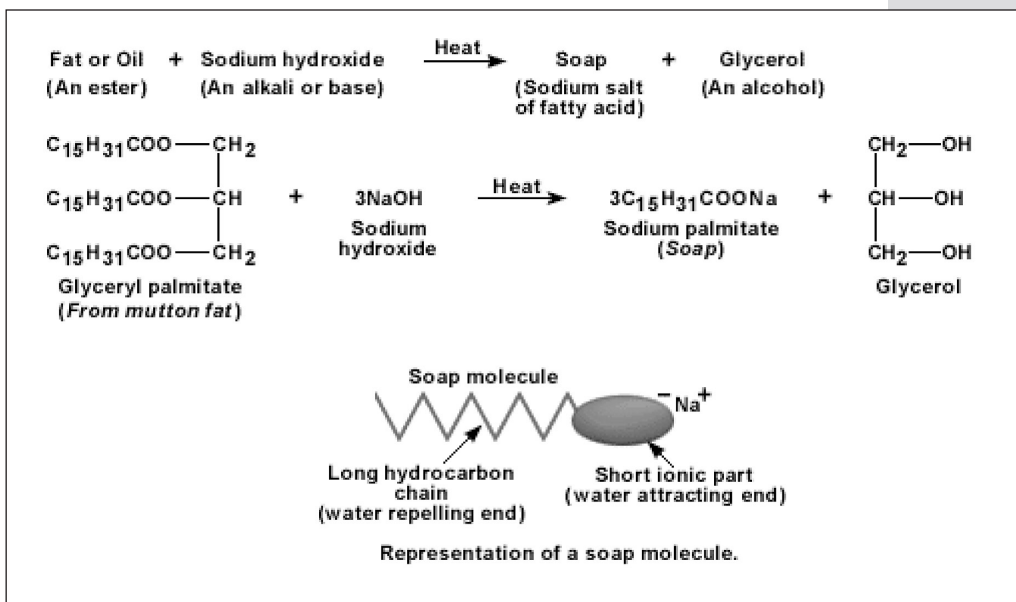
# Everyday chemistry

## 4e Surfactants and detergents

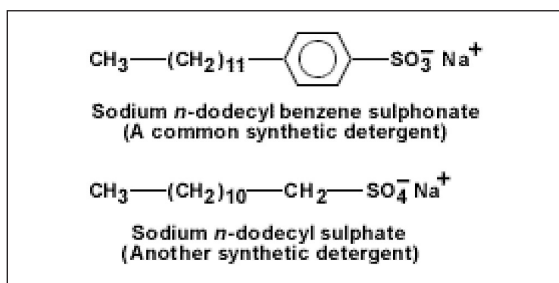
Surfactants and detergents have had a long history.

- 1 a Investigate the history of detergents beginning with the first soaps and continuing to the current day of biodegradable detergents.
- b Construct a timeline to represent this history.
- c Describe the differences in the chemical structure of soap and detergent shown below.

Heating fat along with NaOH makes soap. Sodium salt of a fatty acid and an alcohol is formed. This process is called saponification. One such saponification reaction is shown below.



Synthetic detergents have long chain molecules such as sodium *n*-dodecyl benzene sulphonate and sodium *n*-dodecyl sulphate. The chemical structure of these molecules is shown below.



An article similar to the one below appeared in the local papers during 1965 when the first biodegradable detergents were being developed. Read the article and answer the following questions.

### LOCAL FOAM

Alarms have been raised in regard to the apparent increase in foam that has been spotted in the local waterways. Local scientists have been put to the test.

It has been found that the detergents that have been washed down into the waterways have a very long breakdown rate and therefore the foam seems to be accumulating.

A new detergent, called “Environ-first”, has been developed, which uses micro-organisms to break down the detergent in a matter of days.

Traditional detergents have branched hydrocarbon chains which produce the long-lasting foam. This new detergent, however, contains unbranched hydrocarbon chains which can be broken down by the micro-organisms.

Scientists are asking for the community’s support in changing detergents to save our waterways.

- 2 a Do you ever see foam or froth in local waterways?
  - b Why would ‘alarms be raised’ about the foam in the waterways?
  - c What happens to the waste detergent water from your washing-up sink or the soapy water from your bath or shower?
  - d Could this affect the local waterways? Explain.
- 3 In the article above, the suggested solution to the foam was the new detergent. Explain how this detergent solved the problem.
  - 4 How could community support help improve the waterways?

## 4f Emulsions

### MAYONNAISE vs SALAD DRESSING ACTIVITY

In this activity you are going to make salad dressing and mayonnaise and compare the two substances.

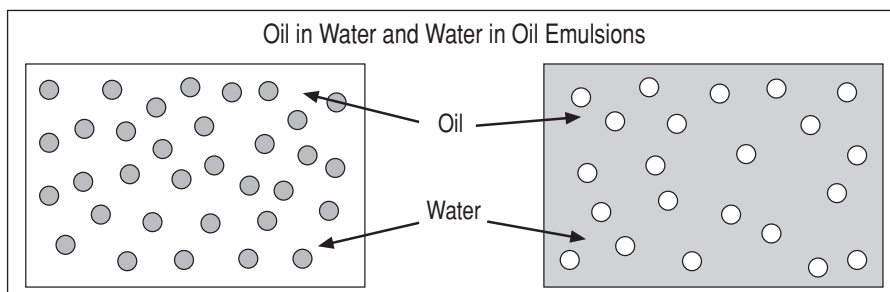
#### Salad dressing

Take a small beaker or cup (if you intend on using the salad dressing), add 1 teaspoon of oil (olive oil is always nice!) then add about 3 tablespoons of vinegar, a pinch of salt, pepper and mustard to taste. Stir the ingredients together and observe what happens.

#### Mayonnaise

In a beaker or small bowl (if you intend to consume your mayonnaise), take two egg yolks, a pinch of pepper,  $\frac{1}{2}$  teaspoon of salt and a  $\frac{1}{4}$  teaspoon of mustard. Mix well. Measure 6 tablespoons of oil and add a few drops at a time while continuing to stir. Measure 2 tablespoons of vinegar and again add a few drops at a time while continuing to stir. Look at the difference between the two substances.

Look at the difference between the salad dressing and the mayonnaise. The only difference in ingredients between the salad dressing and the mayonnaise is the egg yolk. The egg yolk is an emulsifier. It works by allowing the oil and the vinegar to mix within one another by reducing the difference in the surface tension. These two substances are called 'phases'.



There are two main types of emulsions: oil in water (O/W) and water in oil (W/O). By sight there does not appear to be a difference. An example of an oil in water emulsion is moisturiser or cold cream and an example of a water in oil emulsion is sunscreen.

### O/W VS W/O ACTIVITY

Take a small amount of moisturiser and a small amount of sunscreen and rub one onto the back of one hand and the other on the other hand. One should give a warm effect and the other should feel cold.

- 1**
  - a** Which one feels cold and which one feels warm?
  - b** Suggest reasons why one feels cool and the other warm.
- 2** Write a simple conclusion for this activity, stating which type of emulsion would be called the 'warm emulsion' and which gives a cooling effect.

There are two tests that can be performed to confirm whether an emulsion is oil in water or water in oil. One involves using water-based dyes and the other is performed by testing how conductive each is.

- 3** Research these methods and write a procedure for each one.

# Everyday chemistry

## 4g Vitamins

Vitamins are required in small amounts to assist in metabolic activity.

**1** The table below is incomplete. Fill in the blanks using the Phrase Bank.

### Phrase bank

- Found in a wide range of foods
- Healthy nervous system, assists with chemical changes in respiration
- Increased clotting time
- Increases immune resistance, healthy skin, assists in maintaining lining of arteries and veins
- Liver, meats, grains and legumes
- Antioxidant, important in the formation of red blood cells
- Scurvy (easy bruising and bleeding gums)
- Sunlight

| Vitamin                                   | Source                                               | Function                                                                    | Deficiency effect                                           |
|-------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>A</b><br><b>Retinol</b>                | Milk, cheese<br>Margarine<br>Green vegetables        | Assists in proper vision                                                    | Night blindness,<br>Permanent blindness                     |
| <b>B<sub>1</sub></b><br><b>Thiamine</b>   | Pork<br>Whole grains<br>Legumes                      | Removal of carbon dioxide                                                   | Beri-beri<br>Heart failure                                  |
| <b>B<sub>2</sub></b><br><b>Riboflavin</b> | <b>a</b>                                             | Important in maintaining healthy eyes and skin.<br>Helps protein reactions. | Lips become reddened.<br>Cracks develop at corner of mouth. |
| <b>B<sub>3</sub></b><br><b>Niacin</b>     | <b>b</b>                                             | <b>d</b>                                                                    | Skin and intestinal lesions, nervous and mental disorders   |
| <b>C</b><br><b>Ascorbic Acid</b>          | Fresh fruits<br>Green vegetables                     | <b>e</b>                                                                    | <b>g</b>                                                    |
| <b>D</b>                                  | <b>c</b>                                             | Promotes bone growth.<br>Increases the amount of calcium absorption.        | Rickets                                                     |
| <b>E</b><br><b>Tocopherol</b>             | Green leafy vegetables<br>Margarine                  | <b>f</b>                                                                    | Anaemia                                                     |
| <b>K</b><br><b>Phylloquinone</b>          | Green leafy vegetables<br>Cereals<br>Fruits<br>Meats | Blood clotting                                                              | <b>h</b>                                                    |

The following sites will help you with the questions on this page:

[http://www.herron.com.au/healthcare/vitamins\\_guide.html](http://www.herron.com.au/healthcare/vitamins_guide.html)  
[http://www.blackmores.com.au/products/products\\_A-Z.asp](http://www.blackmores.com.au/products/products_A-Z.asp)  
<http://www.kellogg.com.au/DisplayPage.asp?PageID=322&sectionid=1>  
<http://www.betterhealth.vic.gov.au/bhcv2/bhcsite.nsf?open>

**2** List each of the vitamins from the table as being either fat soluble or water soluble.

**3**

- What happens if we take in too much (excess) fat-soluble vitamin?
- Explain why fat-soluble vitamins can be toxic if supplements are taken over a long period of time.
- What happens if we take in too much (excess) water-soluble vitamin?
- Explain why water-soluble vitamins are not toxic if supplements are taken over a long period of time.
- If poisoning due to excessive vitamin intake was to occur, what would you do?

**4** Explain why we need to ensure that we have adequate levels of vitamins and plants do not.

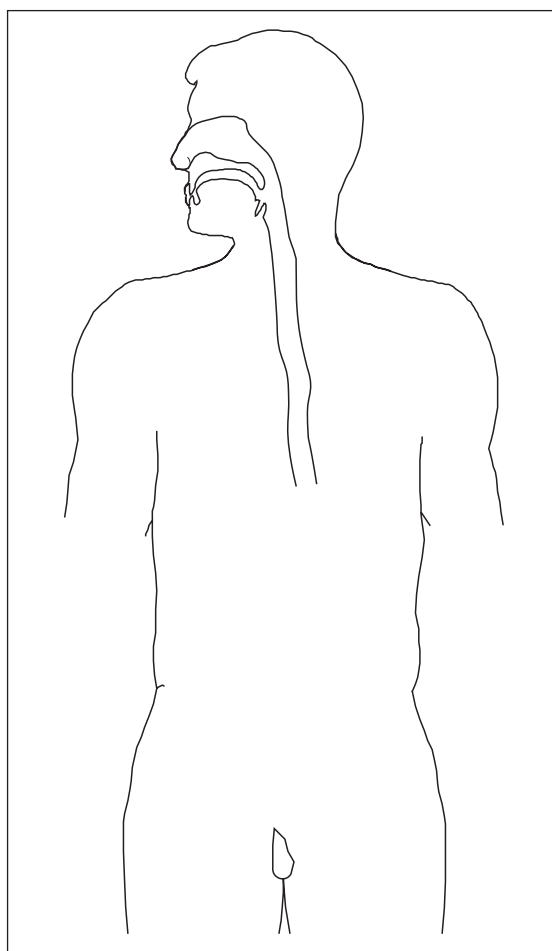
**5** For each vitamin, find out what the recommended daily intake (RDI) is and list it in a table.

## 4h Drugs

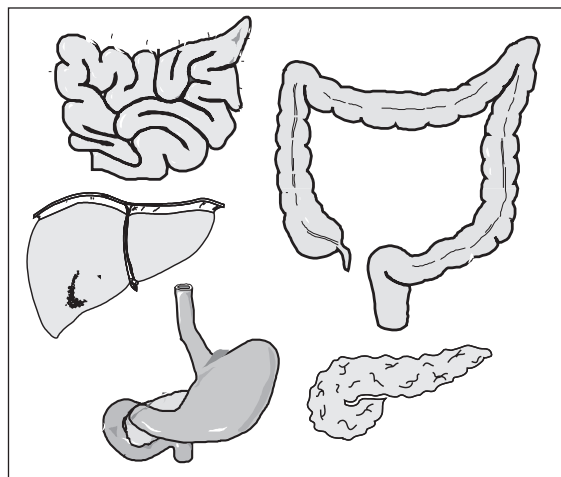
One problem with inventing new drugs is that scientists must be able to ensure that the drug reaches the site in which is meant to take action. Most absorption of foods and nutrients takes place in the small intestine and so for some drugs dissolving must be held off until it reaches the small intestine. To fully understand how new drugs are invented we first need to remind ourselves of the digestive tract.

### 1 Tummy troubles activity

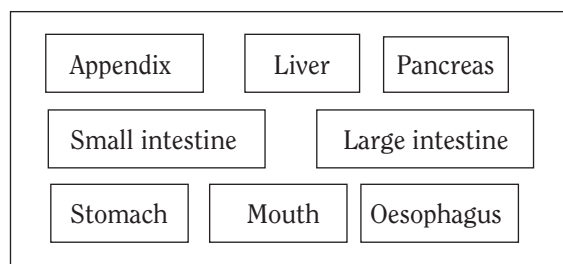
Mike has spilt his guts and it is your job to put him back together! What you need to do is to cut out all of the internal organs below and stick them together in the correct order on the torso. Once you have done this cut out the labels and attach them to with a line to the correct part of the digestive tract and attach to the label the correct function. You can do this individually or as a group.



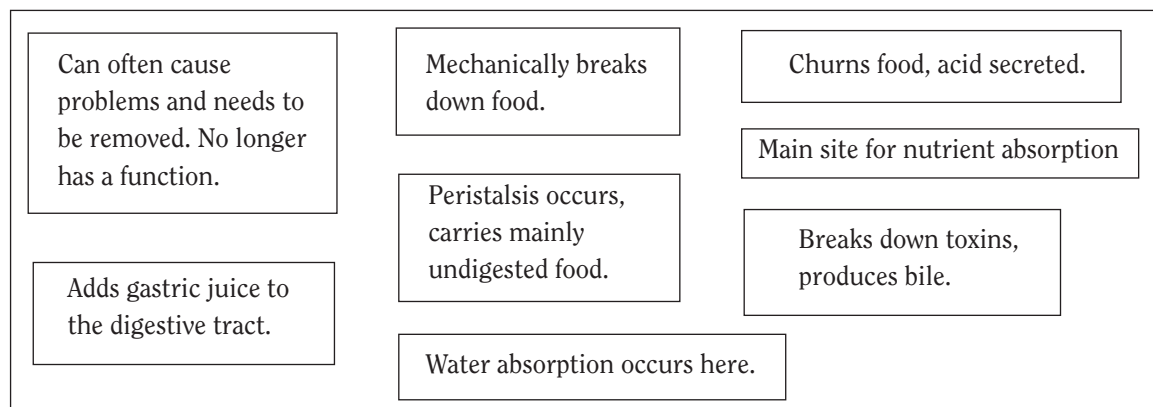
### PARTS



### LABELS



### FUNCTIONS



## 4h Drugs (continued)

The term pH is used as a measure of how acidic something is. A pH of 7 is neutral (like water), below 7 is acidic and above 7 is basic.

The stomach is very acidic and has a pH level between 1 and 3, while the small intestine has a pH between 7 and 9. This information is very important to consider when designing drugs. It is important that the drug is designed to dissolve in the designated part of the body and not react with the chemical reactions that are going on with the general digestion within the body.

Drugs can be administered in many forms, such as:

- Liquid
- Subdermal patches
- Tablets
- Capsules
- Lozenges
- Enteric-coated tablets
- Slow release tablets and capsules

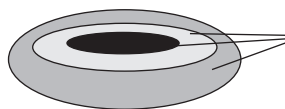
**2** For each type of drug in the list above, write a description of:

- a where in the body it is effective, and
- b what its special requirements are, for instance, does it have a special coating to allow it to only dissolve once it has reached a certain pH?

Slow-release tablets and capsules are designed to work in many different ways.

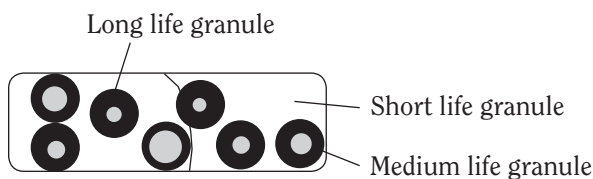
**3** Research this and explain how the different types are effective for the symptoms they are trying to help.

Slow release tablet



Each layer contains an active ingredient that is designed to dissolve at different times.

Slow release capsule



Each granule dissolves at a different time.

### ASPIRIN ACTIVITY

Aspirin belongs to a group of chemicals called salicylates. These chemicals were first discovered thousands of years ago. Hippocrates (founder of the Hippocratic Oath that doctors use) wrote about a 'bitter powder extracted from willow bark that could ease aches and pains and reduce fevers' in 400 BCE. We now call this chemical salicin. Salicin can be changed by the body into salicylic acid. For many years salicylic acid (made from salicin for the first time by Italian chemist Raffaele Piria in 1838) and closely related chemicals were used to treat pain and swelling in various diseases.

Aspirin can come in many forms, such as powder, tablets, capsules and enteric-coated tablets.

**4** Your task is to design an experiment that would allow you to test the solubility of each of these types of tablets in various solvents. You will be allowed to use the following substances in your experimental design.

- Hydrochloric acid (HCl) with a pH between 1 and 3
- Alkaline solution with a pH of 8 or 9
- Water
- Beakers
- Stopwatch
- Aspirin in the following forms: powder, tablets, capsules and enteric-coated tablets.

- a For your experiment write a procedure you are going to follow.
- b Draw a diagram of the set-up you intend to use.
- c Draw up a table in which to record your results.
- d Write a hypothesis on what you think will happen in the case of each type of medication.
- e What is an enteric-coated aspirin?
- f Research the use of enteric-coated aspirin and explain why they are on the market.

## 4i Research Project

In this topic a number of websites have been listed. There are many other sources of environmental information. In particular, the websites for the various commercial drug companies are very useful.

### ABC Health Report

<http://www.abc.net.au/rn/talks/8.30/helthrpt/stories/s357339.htm>

### Aspirin Adventures

<http://www.chemheritage.org/EducationalServices/pharm/asp/asp180.htm>

### Global Health and Fitness

<http://www.globalhealthandfitness.com/aspirin.htm>

### How Stuff Works

<http://www.howstuffworks.com/aspirin1.htm>

### Nobel Prize Internet Archive

<http://almaz.com/nobel/medicine/>

### Salicylates in Foods

[http://users.bigpond.net.au/allergydietitian/fi/FI\\_sal.html](http://users.bigpond.net.au/allergydietitian/fi/FI_sal.html)

You have been told that you should take one aspirin a day, but have also been informed that taking aspirin can often give you stomach ulcers or stomach upsets because they dissolve either in the stomach or before reaching the stomach.

**Task** To determine the best use or dose of aspirin to help reduce pain and fever but also prevent stomach upsets. Your group report will present the unanimous response and will be supported by correctly referenced resources. Your report may include audiovisual aids, such as a computer slide show, verbal presentations and/or poster displays.

## Investigation - pH of household substances and skin

### Part A

Collect a wide range of substances that would be used around the house, eg shampoo, car wash, soap. Form a table similar to the one below:

| Household substance | pH |
|---------------------|----|
| Car wash            |    |
| Shampoo             |    |
| Bleach              |    |
| Aftershave          |    |
| Lipstick            |    |

Using universal indicator paper, test each of the household substances and determine its pH.

### Part B

Dampen another piece of universal indicator paper with distilled water. Place this piece on your wrist and measure the pH of your skin. Repeat this for your forehead and shoulder. Compare your results by testing at least two other people. Construct a table that would be suitable to show these results.

### Part C

From the results that you have obtained from parts A and B, make a conclusion that compares the pH of the skin with that of the household substances.

### Part D

The skin is the body's largest organ. Using the internet or book sources, draw and label a diagram of the skin. Label the diagram with the following parts:

- Hair
- Adipose cells
- Sebaceous gland
- Epidermis
- Sweat gland
- Dermis

# Suggested answers

## 1a Environment Crossword

Across: 1 US 2 gas 9 biodiversity 12 sea 17 environment 18 ant 19 car 20 oil 21 feral 22 land

Down: 1 UN 2 global 3 salinity 4 forest 5 coast 6 Kyoto 7 native 8 population 10 vegetation 11 pollution 13 chemical 14 species 15 petrol 16 energy

## 1b What is the Australian Environment Like?

**1** Answers will vary but include aspects of climate, vegetation and human involvement.

**2 a** water, air, soil, noise **b** Answers will vary but may include waste water, car exhaust, heat loss from heaters and air conditioners. **c** Answers will vary but could include detergents from washing machines going into waste water channels and into rivers, creeks or the sea; car exhaust adds to air pollution levels.

**3** There were very few Aborigines and they were nomadic. Moving on gave the land time to recover and simple living meant there were no machines or manufactured materials such as plastic to create any pollution.

**4** Modern Australia has paved surfaces, fewer trees, more introduced plant and animal species, more grazing land, chemical pollution, light pollution and fewer uncontrolled bushfires.

**5 a** As cities become crowded, more waste is produced and has to be disposed of. Because the waste is removed, people do not realise the amount of waste they produce.

**b** Answers will vary.

**6 a** dry, windy **b** Removal of trees, over grazing, use of artificial fertilisers rather than manure or mulch. **c** Plant trees, maintain areas of grass in the paddock, do not overstock, use natural fertilisers.

**7 a & b** Answers will vary. **c** Effects include waste water adding nutrients to rivers, cars adding chemicals to the air.

## 1c How is Australia's Environment Changing?

**1 a & b** A *predator* (eg dingo) is an animal that attacks and eats smaller animals. *Parasites* (eg mistletoe plant, lice) live on another animal or plant, using it as a source of food or energy without killing it. *Prey* (eg rabbits) are the animals hunted by predators. *Competition* occurs when more than one type of animal or plant needs the same resource (eg sheep and kangaroos both need grazing land). *Erosion* is the removal of dirt and soil from an area by wind or water (eg when mud flows down to the sea in a flood). *Resources* are all things used by organisms for their benefit (eg air, water, food, shelter).

**2** eg: Rabbits eat the grass roots allowing erosion to take place. Cane toads are poisonous and kill native predators who try to eat them.

**3 a & b** Answers will vary.

**4 a** The ability of an animal or plant to produce offspring.

**b** If animals are infertile they will die out rather than produce a new generation. **c** Radiation or chemical baits.

**d** Answers will vary.

**5** Answers will vary, but could include changes in bird and animal populations, stock losses, rainfall or river heights.

## 1d What Makes Up the Environment?

**1 a** A component is something that makes up or is a part of the whole; so the individual parts that make up the environment.

**b** Air pollution: chemicals or particles added to the air;

Biodiversity: the range of different types of living things in a community; Coastal degradation: the gradual destruction of land in a coastal area; Feral plants and animals: domestic pets or garden specimens which have gone wild in the natural bush; Greenhouse gas production: the amount of CO<sub>2</sub> and other gases which, when released, trap heat in the atmosphere; Native vegetation: plants originally found in the area; Population growth: the change in the number of individuals living in a specific area; River systems: creeks, rivers and catchment areas for these waterways; Salinity: a measure of the amount of salt (NaCl); Urbanisation: the build-up of human habitation and associated structures such as roads, schools and factories.

**c** All components are closely connected and directly affect the type and number of plants and animals found in a particular area.

**2 a** 300 years ago there was no urbanisation, the only feral animal was the dingo, population growth was low and pollution was small and localised. **b & c** Answers will vary; changes are due to population growth, industrialisation and urbanisation.

**3 a** 40 000 **b** 103 000 **c** 5000 **d** SO<sub>2</sub> emission increased to a maximum in 1992 then gradually reduced up until 1999 when it dropped suddenly. **e** Industrial activity increased from 1982 to 1992 then reduced and eventually stopped in 1999.

**4 a** steelworks **b** It will reduce the amount of SO<sub>2</sub>. **c** Lack of local product, lack of employment. **d** Answers will vary but relate to loss of income and job opportunities weighed against the environmental benefits.

## 1e Environmental Change and Development

**1 a** Electrical items, computers, machinery, industrial products. **b** Any activity which uses modern technology. **c** We live more removed from the environment. We don't rely on the environment as much, so are less concerned if we damage it.

**d** Modern Australians damage the environment to a much greater extent than early settlers or Aborigines.

**2 a & b** Answers will vary.

**3 a** Transport of raw materials, transport of products, proximity to markets, changes in demand for product or cost of raw materials, cost of city space. **b** Same as for city, but the amount of land used is greater and the distance to markets is greater. There may also be differences in cost of transport.

**4 b** Farm soil is more acidic due to artificial fertiliser.

**5 a** Advantages: improving crop raes; disadvantages: causing poor growth in natives on edges of paddocks, and killing native fauna. **b** Low pH water could acidify waterway killing native flora downstream.

## 1f Air Pollution

**1 a** Breathing difficulties, skin and eye irritation, dirty windows, cars etc. **b** Smokers, traffic, factories and homes. **c** Answers will vary **d** Could include reduced traffic, anti-smoking programs, filters on factory outlets.

**2 a** 6.5 W and 3.2 S for 1982 and 1.2 W and 2.2 S for 202

**b** Strong reduction to 1989 for winter coefficient then reasonably constant. Summer shows increases in bushfires. **c** Paper is collected for recycling and leaves are often recycled in garden compost. **d** Answers will vary.

**3 a** Dust or particulate pollution. **b** Burning of rubbish, household fireplaces, and backburning in the bush to reduce summer fires. **c** Reduction in particulates.

**4 a** Car engines **b** Greater use of public transport, car-pooling, walking or riding a bicycle whenever possible.

## 1g Air Pollution from Cars

**1 a** Answers will vary. **b** Advantages include convenience, time and speed; disadvantages are cost and pollution.

**2 a** 51 in 1988; 47 in 1991 **b** World Expo 88 was held in Brisbane, so people from all over Australia drove large distances to attend. **c** 1 billion

**3 a** 63 billion, 72 billion **b** Will increase to 80 billion.

**4 a** Answers will vary. **b** Advantages include fitness, lack of pollution, cost; disadvantages include safety, convenience. **c** & **d** Answers will vary.

## 1h Looking After Our Water Resources

**1 a** Drinking, washing, cooking. **b** Human waste, washing water, detergents, food wastes. **c** Household waste, road runoff, chemicals from industry, oil and wastes from cars. **d** Fish and other aquatic organisms, frogs, plants that inhabit the banks and animals that live off aquatic organisms.

**2 a, b** & **c** Answers will vary.

**3 a** Answers will vary. **b** Jul - Sep

**4 a** Reduce litter in the streets by providing sufficient bins; put grates or meshing over drains. **b** Answers will vary but include reduction in initial pollution and increases in early intervention to remove pollutants from the water.

## 1i Water Pollution

**1 a** Answers will vary. **b** Prevention would include boiling water before drinking or rinsing food.

**2 a** Something contaminated by animal or human waste has been ingested. **b** Pollution after treating or a fault in the water system. **c** An organism that lives off another without killing it.

**3** Chemical, battery, manufacturing, electrical **b** Untreated water escaping from the manufacturing process or deliberate dumping of waste water to cut costs. **c** They are more dense

than many other metals (mass per unit volume).

**4 a** Cars, mowers, boats, any electrical machinery such as washing machines, driers and garden tools; also cooking oil. **b** Dropping on the road, waste not disposed of correctly, leeching through soil, cooking oil rinsed down drains. **c** It does not allow oxygen to pass from the air into the water.

**5 a** Grocery and household shopping. **b** Take your own bag to the shops; use cardboard boxes that can be recycled.

**c** They choke marine mammals and can trap crustaceans and fish.

**6 a** It gives everyone the same set of standards to maintain.

**b** & **c** Answers will vary but include heightened awareness of environmental issues, reduction in pollution and increase in air and water quality.

## 2a Australian Wildlife Crossword

*Across:* 2 paw 3 back 5 frog 7 crocodile 8 bill 10 wattle 11 galah 14 rosella 15 platypus 16 gum 17 heath 18 lyre 20 cod 21 kangaroo

*Down:* 1 koala 2 parrot 4 alligator 6 mouse 8 bat 9 echidna 10 waratah 12 frill neck 13 marsupial 19 emu

## 2b Coastal Degradation

**1 a, b** & **d** Answers will vary. **c** Built up areas where people live and work.

**2 a** Beachfront development, black sand mining, house building, farming. **b** Answers will vary but in black sand mining, sand is removed, minerals taken then the sand is replaced. **c** Disruption to vegetation; natural structure of coastline altered; vegetation removed, wildlife destroyed etc.

**3 a** NT **b** 12 060 km<sup>2</sup> **c** 16%

**4 a** Erosion can remove soil or sand from underneath the building causing its destruction. **b** It moves and becomes larger and smaller in a seasonal cycle. **c** Houses cannot be built to change with the dune's natural cycle. **d** Deep sand is not protected and supported by the roots of sand grasses.

## 2c Habitat Destruction

**1 a** Erosion could destroy its habitat forcing it to move or even killing it. **b** Urbanisation causes destruction of forest habitats and supports the type of species which prey on the bettong or compete with it. **c** No, because of the effect of rabbits, foxes, cattle and sheep. **d** Maintaining the forest that exists and building corridors across farm land.

**2** Answers will vary.

**3 a** 1994 **b** 27 000 tonnes **c** 230 tonnes **d** The salt water fish industry is closer to population centres and there is more water in which to fish. **e** Reduction in catch. **f** Overfishing.

**4** Lower fish stocks could lead to reduction in food source and removal of species may encourage growth of non-native species, which will further destroy river system.

## 2d Sewage Pollution

**1 a** BOD is quantity of dissolved oxygen used by organic matter. **b** Increase processing in sewage system. **c** Make sure waste

water has all chemicals removed before it is released. **d** Answers will vary.

**2 a** Answers will vary. **b** 80%, 20%, 15%, 5%. **c** In the flat area near the river or sea. Sewage would then flow into the sea.

**3 a** Phosphorus salts and faecal coliform. **b** Faecal coliform is found if faecal matter (animal or human) has contaminated the water. **c** Yes, as it may contain disease-causing bacteria.

## 2e Chemical Contamination

**1 a** Tidal region where the river enters the sea. **b** Salt water washes in while fresh water washes out.

**2 a** At the bottom of sandy estuaries. **b** Animal living on biological matter that has sunk to the bottom; eg prawn. **c** Meat, usually smaller animals; eg shark.

**3 a** detritus feeders **b** top carnivore level **c** detritus feeders **d** top carnivore level

**4 a** Each carnivore eats many smaller animals which have fed on many detritus feeders, all of them containing the pollutant. **b** → In each case the arrow direction is from crustacean to fish.

## 2f Feral Animals

**1** *Urbanisation*: growth of factories and homes and associated infrastructure eg Gold Coast Qld; *feral*: animals, usually domestic cats and dogs, released into the wild; *domestic*: relates to household pets such as cats and dogs; *eucalyptus*: genus of Australian native plant which includes gum trees; *predators*: hunt other animals for food eg snakes; *carnivore*: meat-eating animals, eg lions; *suburban*: small part of an urban area.

**2 a** Most native birds, small mammals and reptiles.

**b** Any non-native animals, eg cats, dogs, cane toad, fire ant, rabbit, fox. **c** Answers will vary but the most well known is the rabbit which causes major damage to farming land.

## 2g Feral Animal Control Program

**1** Answers will vary, eg common pest is the cane toad which poisons native animals that try to eat it.

**2 a** Native: mountain galaxias, eels, freshwater catfish, Australian smelt, Australian bass, freshwater mullet, gudgeons, bony herring, golden perch. Introduced: brown trout, rainbow trout, carp, western carp, redfin perch, gambusia. **b** The carnivores eat native fish and are usually more aggressive so obtain more of the scarce resources.

**3 a** Not to put them back into the water live and not to move them from place to place. **b** So they don't reproduce and spread to other waterways. **c** Good fighting fish and good eating.

**4 a** National Feral Animal Control Program **b** Information about lifecycle, threat to native wildlife and best practice in control. **c** One can see where species are present and look at the different environments to calculate limits to the spread of the animal.

## 2h Global Warming

**1 a** Movement of birds and butterflies. **b** 2° to 3° **c** Ice caps melting, increasing sea level, weather and climate changes.

**2 a** Coastal dwellers might consider rising sea level and country responses will relate to climate changes and drought or rain.

**b** Answers will vary.

**3 b** The diagram shows the CO<sub>2</sub> released from the city causing an increase in temperature. **c** CO<sub>2</sub>

**4 a** Combustion reactions release CO<sub>2</sub> and the others are released from a range of different chemical reactions.

**b** Answers will vary but include reusing and more effective use of material we have, filters to reduce release of chemicals and less energy waste.

## 2i Greenhouse Effect

**1 a** CO<sub>2</sub> and other gases. **b** Keeps the day cooler and night warmer. **c** Increasing temperature over the earth.

**2 a** carbon dioxide **b** Combustion of fuels. **c** 0.035%

**d** 0.027% **e** Production of gases from burning fuels for industry.

**3 a** US **b** 6.6 million tonnes **c** 6.5 million tonnes **d** They use less energy in transport and their energy is more efficiently produced.

**4** Answers will vary.

## 3a Radiation Crossword

*Across*: 1 D<sub>2</sub>O 2 nuclear 7 E = mc<sup>2</sup> 8 Rn 10 hydrogen 11 power 13 Lr 15 fission 16 Am 17 heat 18 Cd 19 neutron 21 alpha 24 At 25 CAT 26 gamma 27 Bohr 28 uranium 29 atom

*Down*: 2 nucleus 3 low 4 Rd 5 sun 6 xray 7 EMR 9 deuterium 11 alpha 12 proton 14 radiation 20 tritium 22 Pd 23 Ac 26 gun

## 3b What is Radiation?

**1 a** Positive nucleus with protons and neutrons surrounded by negative electrons. **b** We have a research reactor.

**2 a** negative **b** In the nucleus. **c** empty space

**3 a** Large amount of energy and no greenhouse gases are produced. **b** & **d** Answers will vary. **c** Radiation and dangerous waste products.

**4** They are lost from the system as radiation.

**5 a** Yes **b** It can cause our skin to burn and glare can harm sensitive eyes. **c** Wear protective clothing, hat, sunglasses and sunscreen. **d** Different types of radiation affect different parts of the body.

## 3c Radiation and Health

**1 a** They can observe internal structures. **b** Broken neck or an accident to the head or neck. **c** Reduce exposure to the rest of the body and to technicians. **d** Shouldn't worry as it is a very small, isolated part of the body. **e** May cause more concern as it is a larger bone and located in the middle of the body.

**2 a** Controls and interprets x-rays. **b** Yes, as they could be exposed to regular doses of radiation. **c** Yes because the effect might be cumulative.

**3 a** electromagnetic radiation **b** Electronic equipment with electrical current. **c** It is used for broadcasting and receiving information.

**4 a** Unborn baby. **b** Radiation may affect the growth and development of an embryo and could cause mutations in the dividing cells. **c** Answers will vary, but any radiation may affect embryonic development. **d** & **e** Answers will vary.

### 3d Ionising Radiation

**1 a** 2 protons and 2 neutrons **b** electron **c** + **d** Many metres. **e** It is electromagnetic radiation.

**2 a** How far radiation can go into a material. **b** Particles are larger and hit the particles in a material therefore slowing down and stopping.

**3 a** (information from table) **b** Measuring thicknesses of materials, tracing pipes or flows, killing cancer cells etc. **c** 7363 times **d** Much greater affect on the surface cells of the body.

**4 a** They are deflected. **b** Different charges are differentially affected by magnetic fields. **c** Electrons the most, alpha particles least; gamma rays are not deflected. The amount of deflection is determined by the charge/mass ratio.

**5 a** skin **b** Answers will vary.

### 3e Continuing Nuclear Radiation

**1 a** 15 hours **b** 15 hours **c** 15 hours

**2 a** The mass of Na 24 has reduced by a factor of 16. **b** Continues to drop at the same rate. **c** When there is no Na 24 left.

**3 a** Almost 12.5 years for hydrogen 3; 5 years for cobalt 60. **b** Little more than 6 t.

**4 a** 28 years **b** beta **c** 55 s

**5 a** 24 000 years, then 12.5 t of Pu 239 and 12.5 t of some other radioactive material. **b** Answers will vary.

### 3f Non-ionising Radiation

**1 a** 100 m **b** am radio **c**  $10^{-7}$  m **d** radio receivers, skin, eyes, photographic detectors **e**  $10^{10}$  to  $10^{12}$ ,  $10^{15}$ ,  $10^{12}$ ,  $10^{20}$  **f** Hard x-rays and gamma rays.

**2 a** 3.23 m **b** & **c** Answers will vary. **c**  $10^9$  Hz

### 3g Safety and Non-ionising Radiation

**1 a** 10 million, 30 million **b** 20 million **c** 1 year

**2 b** Answers will vary but could relate to comfort with technology and price of phones and calls.

**3 a** The DNA is replicating. **b** The nuclear material is spread over a larger volume of the cell. **c** Answers will vary but could include using hands free technology and reducing mobile use. Obviously stay out of the sun and DO NOT smoke.

**4 a** Wavelength, frequency and therefore energy. Yes because type of energy determines the affect on cells. **b** Answers will vary but one has to weigh up the risk with advantages of the use of any item or product. **c** Generally yes; we do not know yet what a 'safe' level is yet.

**5 a** Light is EMR which we see and IR can be felt in the form of heat. Other radiation can be felt if it 'burns' the skin. **b** Radio receivers, antenna, electrochemical reactions etc.

### 3h Extra-low Frequency

**1 a** Gamma rays and x-rays at one end, ELF waves at the other. **b** Gamma rays have high frequency, short wavelength and high energy. ELF waves have low frequency, long wavelength and lower energy. **c** Extra Low Frequency **d** 500 kV **e** 240 V

**2 a** EMR of ELF **b** Cause fluorescent tubes to glow, cause hair

to stand on end and produce charges on parked cars.

**c** Concerns include increases in cancers and birth defects due to cellular destruction.

**4 a** 50 Hz **b** Answers will vary. **c** Yes

**5** Answers will vary.

### 3i Health Concerns and the Community

**2 a** Concerns include proximity to Tully State High School, Tully Airfields and the Tully skydiving industry. It could also increase costs to cane growers, result in a drop in property values and have a great visual impact as well as affecting wildlife and the health of residents. **b** & **c** Answers will vary.

**3** Answers will vary but should include a testing of the veracity of claims and the searching of as many sources as possible.

### 4a Chemistry Crossword

*Across:* 1 emulsifier 4 tyndall 5 water 9 oil in water 10 soap 11 solvent 12 surface tension 13 colloids

*Down:* 2 mayonnaise 3 adhesion 6 polar 7 suspensions 8 surfactants

### 4b Solutions, Suspensions and Colloids

**1** *atom*: smallest particle that makes up matter; *molecule*: consists of more than one atom; *particle*: tiny pieces of matter; *salt*: ionic compound that contains a metal and non-metal atom(s); *gold*: metal element which is highly conductive.

**2** *Solution*: transparent, made by dissolving a solute into a solvent. *Suspension*: mixture of two substances, one of which is finely divided and dispersed in the other (the more dense the lower the layer). *Colloids*: contain small particles that are suspended and do not separate out after long periods of time.

**3** Salt water – solution; Muddy water – suspension; Sunscreen – colloid; Mayonnaise – colloid; Milk – colloid; Tea or coffee – solution; Soft-drink – solution; Salad dressing – suspension; Oil and water – suspension; Sponge cake – colloid.

**4** Answers will vary.

**5 a** Colloid as it allows for the pigment to be suspended for a long period of time. **b** colloid **c** Pigment is suspended.

**6 a** Used for painting over the ink to prevent visibility. **b** colloid

### 4c Tyndall's Effect

**1** You could place a small amount of each solution into a test tube and shine a laser light through. If the light travels through then it is either a solution or a suspension. If the light refracts through the solution and does not go through then it is a colloid. To determine between the solution and the suspension you would allow the substance to sit for a while and if layers form then it would be a suspension, otherwise a solution.

**2** Tyndall's effect works by allowing light to travel through a substance and observing the effects of whether the light is able to travel through or whether the light refracts.

**3 a** A - suspension, B - colloid, C - solution.

**4** Light refracts when it hits the particles of either a suspension (A) if mixed or a colloid (B). In the solution (C), the solute is

fully dissolved and therefore the light travels directly through it.

**6** Particle size: smallest, small, largest

Is light refracted?: no, yes, yes if mixed

Can the particles be filtered out?: no, no, yes

Do particles settle out over time?: no, no, yes

Uniformity: homogenous, homogenous, heterogenous

#### 4d Surfactants and Surface Tension

**1** Answers will vary.

**3** Needle floats on the surface. **4** Needle sinks.

**5** The surfactant surrounds and separates little bubbles of oil for removal.

**6** *Anionic*: negative charged surfactant; *Cationic*: positive charged; *Non-ionic*: non-specific charged; *Amphoteric*: non-charged; *Polar*: charges separated in the molecule; *Non-polar*: no separated charges; *Hydrophilic*: water-loving; *Hydrophobic*: water-hating

**7 a** **A** non-ionic (no charge) **B** anionic (–) **C** amphoteric (+/–) **D** cationic (+) **b** The area with the circle is the hydrophilic end and the tail end is the hydrophobic end.

#### 4e Surfactants and Detergents

**1 a & b** Answers will vary. **c** Show the chemical composition of soap and detergent. Note that the detergent has a very long carbon chain which is easily broken down.

**2 a** Answers will vary. **b** It means that detergents have made their way down to the waterways and that eutrophication of the waterways could develop. **c** The waste water flows down to the water treatment stations or ocean outfall depots and then into our waterways.

**3** The detergent solved the problem as micro-organisms are able to easily break down the unbranched carbon chains.

**4** Awareness of the problem should make people dispose of their waste water thoughtfully and use environmentally-friendly detergent products.

#### 4f Emulsions

**1 a** The oil in water emulsion (moisturiser) will feel cooler than the water in oil emulsion (sunscreen).

**b & 2** In the oil in water emulsion, the water in which the active ingredient is suspended evaporates removing heat, but the water in oil emulsion traps the water and heats the water that is surrounded by oil, giving a warming effect.

**3** Answers will vary.

#### 4g Vitamins

**1 a** Found in a wide range of foods **b** Liver, meats, grains and legumes **c** Sunlight **d** Healthy nervous system, assists with chemical changes in respiration **e** Increases immune resistance, healthy skin, assists in maintaining lining of arteries and veins **f** Antioxidant, important in the formation of red blood cells **g** Scurvy **h** Increased clotting time, which may result in nosebleeds or internal haemorrhaging

**2** Fat-soluble vitamins include: A, D, E and K.

Water-soluble vitamins include: C and B complex.

**3 a** The fat-soluble vitamin absorbs into the body tissue and becomes toxic, poisoning the body. **b** The fat-soluble vitamins can be toxic if taken over time because they may accumulate in the body tissue. **c** If too much water-soluble vitamin is taken it is excreted. **d** The water-soluble vitamins are not toxic as they dissolve in the water and are removed with the excess water when we urinate. **e** Call the poisons information number 131126 and follow directions.

**4** Plants get their nutrients from the soil and can die if they do not have the correct materials. Humans are heterotrophs and must ensure that all nutrient requirements are eaten from their range of foods.

#### 4h Drugs

**1** Mouth: mechanically breaks down food; Oesophagus: peristalsis occurs, carries mainly undigested food; Stomach: churns food, acid secreted; Pancreas: adds gastric juice to digestive tract; Liver: breaks down toxins, produces bile; Appendix: can often cause problems and need to be removed; Large intestine: water absorption occurs here; Small intestine: main site for nutrient absorption.

**2 a & b** Liquid form: effective in the small and large intestine where absorption can occur, no special requirements. Sub-dermal patches: effective in the adipose or fat tissue layer, active ingredient must be fat soluble. Tablets: effectiveness begins in the stomach and small intestine although the starch that the active ingredient is compressed in is broken down in the mouth; no special requirements. Capsules: same as tablet but must wait until the capsule itself has dissolved. Lozenges: effective from mouth; water soluble. Enteric-coated tablets: absorbed in the intestine, need alkaline solution for the coating to be removed. Slow-release tablets and capsules: effective from the stomach on and will dissolve slowly over a period of time.

**4 e** Enteric-coated tablets are used for drugs such as aspirin. They are coated so that the coating will only dissolve once it has come into contact with an alkaline solution such as in the duodenum. Here the coating will be dissolved and the active ingredient can be absorbed. This is effective for elderly people who may have prescribed to take aspirin to thin the blood. Aspirin tablets can have the effect of causing stomach ulcers and this can be reduced via the enteric-coated tablets.

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